The density-environment of radio-selected galaxies

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The density-environment of radio-selected galaxies

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Abstract

We investigate the properties and the environment of radio sources with optical counterpart for the combined VLA-COSMOS 3GHz and MUSE-GTO COSMOS-groups. The multiwavelength character of the COSMOS field offers high quality photometric redshifts, that we will compare with the spectroscopic ones from MUSE, as well as a trustworthy classification of galaxy activity, that we will verify by means of optical diagrams classification.
With the matched catalogue we will explore the nature of radio sources and their close environment (a maximum cylinder of 487kpc of radius in projected distance and $\pm$2000km s^{-1} in depth, when using only spectroscopic redshifts), and compare it with those of the non radio-emitters, only present in the MUSE data set. We will use the obtained results to test the well behaviour of the EAGLE simulation in reproducing the observations, and explore its limitations.
We see no evidence so far for a correlation between SF or AGN radio-emitters with the density environment, but what is observed is a preference for low star-formation rate over star-forming galaxies at high-densities in the EAGLE model, and higher stellar-masses for AGN radio-sources than non radio-emitters. Nevertheless, a larger sample will be needed to confirm these statements.

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1. Introduction

Understanding the density environment of galaxies and the strength with which different properties correlate with it is a fundamental step in the study of galaxy formation and evolution. In the past decade, this field has undergone a significant revolution with the availability of large, multiwavelength surveys, together with the effort made to obtain high quality spectroscopic data.

On the other hand, detetermining if galaxy properties are the result of assembling mechanisms, product of physical processes over a long period (nurture), or if they were established before and conduct the manner in which structures are bounded (nature) is still an open research.

A first approach to this problem relates with several studies that have proven a relation between morphology and environment[1,2], where red early-type elliptical galaxies dominate in higher density regions than blue late-type disk galaxies. At the same time, morphology has also been probed to be related with the star formation history of the source, referring to them as *star-forming* or *passive* galaxies respectively, which suggest that these two elements should depend on their environment in the same way.

While the studies regarding the dependence with the star-forming activity seem to be in clear agreement, the environment of AGN appears to be an open discussion. Miller et al., (2003)[3] claimed that the fraction of galaxies with AGN activity is independent of environment, whereas Kauffmann et al.(2004)[4] suggested that, for a fixed mass, the low-density regions are twice more populated than high-density ones by AGN hosting galaxies.

Where the consensus appears to be more noticeable is regarding the study of radio-loud AGN versus radio-quiet AGN. It was found that the former tend to occupy more massive halos and overdense environments, also having a strong relation with their radio-luminosity. Indeed, radio-

selected AGN within clusters are more powerful than those associated with isolated galaxies[5,6].

Given the above, the main problem encountered while comparing different studies resides in both the classification of galaxy activity and the definition of environment.

When studying the source-type, making a sharp division between star-forming galaxies and AGN can be really hard, as it is already known that both phenomena coexist in a number of galaxies[7]. Supposing that the trigger mechanism that relates them with a particular environment is very different, making assumptions about those composite sources becomes difficult. Moreover, the contamination of one type over the other can lead to contradictory results.

Regarding the definition of neighbours, the bigger issue resides in the accuracy with which redshifts are determined. Most of the studies try to find sky fields where spectroscopic surveys are available, but the amount of high quality spectroscopic observations of large areas is limited. Alternatively, photometric redshifts are also used, but they undergo higher uncertainties.

In this project, we attempt to study the evolution of the density-environment with respect to the distance to the source, which can help to give a better insight on the range of influence of the different galaxy properties. A well defined scale, in this case, is totally necessary for comparing possible divergent behaviours. Therefore, we decided to explore the new potential of optical high-quality spectroscopic observations, which allow a better determination of spectroscopic redshifts, over the neighbouring area of radio-detected sources. We will also explore the possibilities that the EAGLE simulation could open, and test its degree of replication of observational results at the scale of this project.

The paper is structured as followed: Chapter 2 resumes the most important features of the used data. Chapter 3 describes the processes carried out prior to the analysis of the density-environment, such as matching of the employed surveys and the imposed restrictions on the counterpart selection, or the description of the machinery used to determine galaxy properties. In chapter 4 we expose the obtained results regarding the study of redshift and source-type of the galaxies, as well as its environment, to finally compare with simulations. Ultimately, chapter 5 contains the interpretation of our results.

Throughout this paper, we have assumed, for observations, a WMAP9[8] cosmology with $H_0 = 69.3$ km s^{-1} Mpc^{-1} and $\Omega_m = 0.286$.

2. Data

2.1 COSMOS

The Cosmic Evolution Survey (COSMOS)[9] is the name of a field that covers an area of $1.4 \times 1.4\,\mathrm{deg}^2$, located near the celestial equator. This privileged region assures visibility by a large number of astronomical facilities from both hemispheres, making possible extensive multiwavelength ground- and space-based observations.

It contains some of the deepest data for such a large areal coverage, allowing the study of a wide number of topics, such as evolution of Large-Scale Structures, gravitational lensing and Dark Matter distribution or assembly of galaxies at medium and high redshift.

Our study of radio sources in this field is based on the particular area covered by MUSE observations, from where VLA data is also available.

2.1.1 MUSE-GTO COSMOS-groups

MUSE, the Multi-Unit-Spectroscopic Explorer[10], is an instrument installed in the UT4 telescope at the Very Large Telescope (VLT) of the Europear Southern Observatory (ESO), Cerro Paranal, Chile. It is an integral field unit (IFU) that yields spectroscopic measurements for each individual pixel (spaxel) within the field of view (FoV). The final mosaic is a square field of approximately $1 \times 1\,\mathrm{arcmin}^2$, sampled with $0.2''$ spaxels resolution, and covering wavelengths between $4650 - 9300$Å.

The galaxy sample used in the framework of this research is extracted from the 10 galaxy groups, within the COSMOS field, observed with MUSE during the Guaranteed Time Observations (GTO), as part of a program focused on *How did environment affect galaxy evolution over the past 8Gyr?* (PI: T. Contini). They are selected from the zCOSMOS 20k group catalogue[11],

by fulfilment of the following criteria:

> **Redshift range:** `z_group = 0.25 - 0.85 to get [OII]3727, Hb, [OIII]5007 (and Ha up to z=0.4)`
> **Richness:** `groups with at least 5 spectroscopic members`

*Intermediate-z Galaxy Groups Candidates for MUSE. Authors: T. Contini, B. Epinat, and L. Michel-Dansac**

To clarify, the data employed throughout the project does make use of all the galaxies within the FoV (for which astrometry is matched to the HST COSMOS coordinates from COSMOS2015 catalogue[12]), and not only of those ones belonging to the selection-groups. Although the fields are named upon the selected targets.

NOTE: The name (number) of the groups are referred as the group ID in the Knobel v3.01 catalogue (that was changed in a later version).

2.1.2 VLA-COSMOS 3GHz

Although the majority of the analysis carried out has been done using the MUSE-GTO data set, the centre of this research was the study of the environment of radio sources. These targets were selected from the VLA-COSMOS 3GHz Large Project catalogue[13], which contains more that 10000 sources over a 2.6×2.6 deg^2 area, enclosing the full COSMOS field.

The 384h of observations with the *Karl G. Jansky* Very Large Array (VLA) constitute the largest and deepest radio continuum survey at high angular resolution, adding new information to one of the current richest multiwavelength data sets as COSMOS is.

2.2 EAGLE

The Virgo Consortium's Evolution and Assembly of GaLaxies and their Environments (EAGLE)[14,15,16] project consists of a suite of hydrodynamical simulations with the aim of studying the formation and evolution of galaxies and supermassive black holes in a cosmologically representative volume, adopting the cosmological parameters derived by the Planck Collaboration[17].

*Quotation from: MUSE science team wiki. Galaxy-groups-field: Selection criteria

Each simulation has information about the evolution of particles, distributed across five SQL tables, with a total of 29 snapshot outputs related with redshifts between 20 and 0. The SubHalo table was the one used to obtain information about the galaxy properties, linked with the Aperture table, which provides the choice of spherical size of the galaxy.

EAGLE defines galaxies as gravitationally bound subhalos, the resulting substructures identified as overdense regions within halos of friends-of-friends[18] linked dark matter particles. Subhalos (galaxies) that are separated by less than 3(proper-)kpc or the stellar half-mass radius are merged together, to avoid having structures only containing a single massive particle such as SMBH.

Moreover, the position of each galaxy is determined by the location of the particle for which the gravitational potential is minimum, while its stellar-mass is defined as the sum of the masses of all star particles within the chosen aperture. Regarding the black hole mass of each subhalo, it refers to the summed value of all black holes which, for the cases where it is greater than $10^6 M_\odot$, closely approximates the mass of the central supermassive black hole of the galaxy.

The performance of the simulation has been successfully tested to reproduce, among others, present-day stellar mass function of galaxies, correlation of black hole and stellar mass, or properties of intergalactic medium, in both present day and early epochs. This makes EAGLE a powerful machine for understanding the evolution of galaxies and interpreting observational datasets.

3. Matched catalogue and properties determination

3.1 Determination of central sources

3.1.1 Matching MUSE-GTO and VLA 3GHz catalogues

The main goal of the project was to characterise the properties and local environment of the VLA-Radio sources in the COSMOS field. The selection of the studied sources was realised by a cross match, by its position in the sky (RA and DEC with an offset $<1.0''$), of this catalogue with the MUSE-GTO data groups in the COSMOS field, resulting in a total of 22 sources. The rest of VLA sources located in a region for which MUSE observations are still not available were omitted in this study.

Fig 3.1 shows the HST image of two of the studied centres [ID_VLA: 4148 and 2454], as representative for the rest of them, where their VLA (red) and MUSE (green) position are marked. It should be mentioned that the HST images correspond to the ACS v2.0 mosaic[19,20], and were extracted to be aligned in astrometry with the inclination of the MUSE white-images.

On the other hand, it was not always possible to find a MUSE counterpart for each VLA source, even though its position was inside the observed region. We were still interested in studying these galaxies (8 sources in total, in the future usually referred as *extra sources*), therefore, they were stored separately for further analysis. Fig. 3.2 represents the COSMOS-groups (CGR) 24 and 77, where red circles correspond with matched sources and blue circles with the aforementioned *extra sources*.

Having created a combined catalogue with both MUSE and VLA data allowed us to have, for example, information of both spectroscopic and

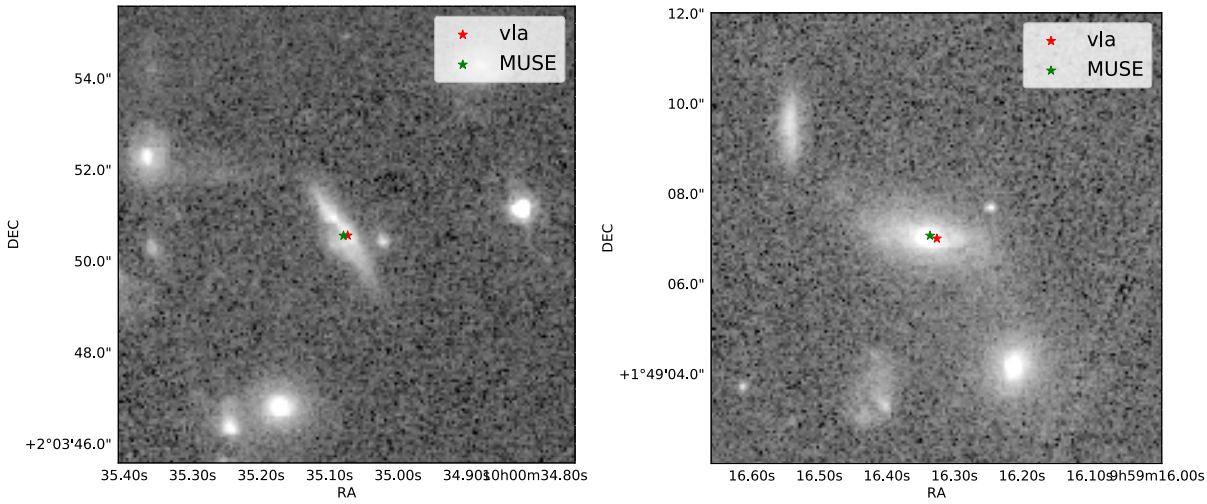


Figure 3.1: *HST image of galaxies [ID_VLA: 4148 and 2454], left and right respectively. The red star is located at the (RA,DEC) regarding VLA3GHz catalogue, while the green star corresponds with the (RA,DEC) according to MUSE-GTO.*

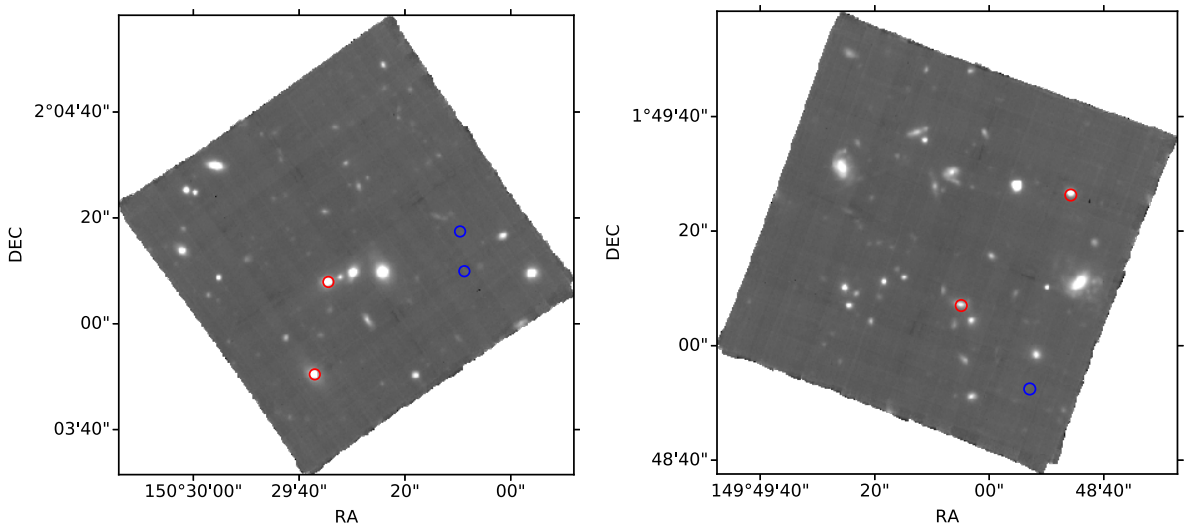


Figure 3.2: *MUSE image of the COSMOS-groups 24 and 77, left and right respectively. Red circles correspond with VLA sources for which was possible to find a MUSE counterpart, while blue circles are those sources that, even though inside the field of view, were not matched with any MUSE counterpart.*

photometric redshift for the majority of the sources, as well as flux lines information or stellar mass[†] (which will be used to classified the nature of the galaxy activity). Together, the cross-matched sources plus those for which it was not possible to find a MUSE counterpart , give us a total of 30 centres for our study.

3.1.2 EAGLE counterparts

Due to the lack of an extended bibliography in the study of local environment of galaxies at the scales of this project (up to $\sim$ 500 kpc), we will compare our results with those obtained by means of the EAGLE hydrodynamical simulation. Therefore, we want to find EAGLE counterparts to our selected central sources.

EAGLE particle data information is accessible via SQL queries, that can be typed in a text box at the database webpage (username and password are needed for access). In addition, a python module to submit queries to the database has been also developed (EAGLESQLTOOLS[‡]), and serves as an interface to access the EAGLE database from outside the webpage.

The two models used (*RefL0100N1504_Subhalo*, *RefL0100N1504_Aperture*) contain respectively the following used information:

- Snapshot number, directly related with the redshift, at which these properties are computed; total instantaneous accretion rate of all black holes [$M_\odot yr^{-1}$]; co-moving position (x,y,z) of the centre of mass [cMpc]; peculiar velocity (x,y,z) in km/s.

- Radius of the aperture used for the mass measurement, taken from the centre of potential [proper-kpc]; star formation rate within the selected aperture (the mass of stars formed per unit time before any mass loss due to winds and supernovae) [$M_\odot yr^{-1}$]; total (particle) mass of all black holes within the aperture [$M_\odot$]; total stellar mass within the aperture [$M_\odot$].

To enable a future full comparison with our observable data, we need to select the centres and environment (neighbours) from the EAGLE simulation that are the most similar to ours. A number of restrictions have been imposed depending on whether we were matching the central sources or its environment:

[†]Computation of stellar-mass realised by: Leindert Boogaard

[‡]It can be downloaded here: http://icc.dur.ac.uk/Eagle/Database/eagleSqlTools.py

- **Counterparts to the radio-sources (centres):** For each centre, up to the top 100 counterparts were selected, for which the conditions were:

  1. The counterpart needed to have a stellar mass close to that of the observed source.

  $$\log_{10} \mathrm{M}_*^{Eagle} = \log_{10} \mathrm{M}_*^{MUSE} \pm 0.1 \tag{3.1}$$

  2. The selected snapshot number corresponding to that one of the closest redshift to the spectroscopic redshift of the central source.
  3. The galaxy activity, although it was not itself a condition for counterpart selection, was later also studied to cross-match sources of same type.
     - **SFG:** The threshold value ($sSFR \equiv \frac{SFR}{M_*^{Eagle}}$) that determines whether a galaxy is star-forming or not, it has been computed to depend on redshift. Galaxies are considered star-forming if

     $$\frac{SFR}{M_*^{Eagle}} > 10^{-11,-10.4,-10.0,-9.6,-9.4,-9.1,-9} \quad \mathrm{yr}^{-1} \tag{3.2}$$

     for redshifts (0, 0.5, 1, 2, 3, 4, < 5)[21] respectively, and do not host an AGN.
     - **AGN:** If the galaxy hosts a SMBH ($\mathrm{M}_{BH} > 10^6 \mathrm{M}_\odot$), its activity level has been based on its Eddington ratio ($\lambda_{Edd} \equiv \frac{\dot{M}_{accr}}{\dot{M}_{Edd}}$)[§], where $\dot{M}_{accr}$ and $\dot{M}_{Edd}$ are the SMBH mass accretion rate and Eddington limit respectively. If the galaxy does not present signatures of star-forming and

     $$\lambda_{Edd} > 10^{-4} \tag{3.3}$$

     it will be considered *active*[22].
     - **Composite:** If it presents both signatures of SFG and AGN.
     - **Error:** Galaxies not showing any particular signature were classified as errors and not taken into account in the source-type restricted study (they are included in the one that does not select by source-type, and will be briefly explore in Sec. 5.3).

[§] $\dot{M}_{Edd} = 3 \times 10^{-8} \left(\frac{0.06}{\eta}\right) \left(\frac{M_{BH}}{M_\odot}\right) M_\odot yr^{-1}$, where we chose and efficiency $\eta = 0.1$

- **Environment (MUSE sources before referred as neighbours):** Close galaxies were analysed for each of the available counterparts individually, following the conditions:

  1. Cut-off in stellar mass $\log_{10} M_*^{Eagle} > 8$, chosen to match the distribution of our sample (see Fig 3.3). Although this represents the total MUSE sample, and not only the galaxies classified as neighbours, we can already appreciate that the majority of them have stellar masses over the chosen limit.

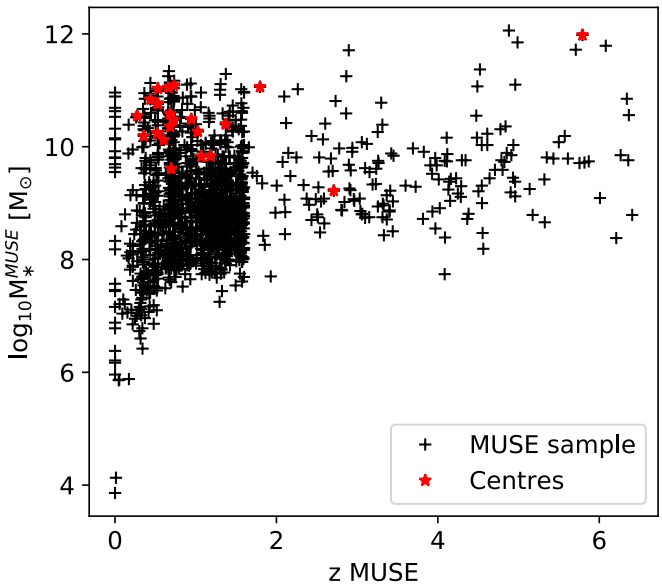


Figure 3.3: *Distribution of stellar mass with respect to redshift. Black crosses correspond to the total MUSE sample (all the sources presented in the 10 COSMOS-groups catalogues analysed), while red stars are the centres for which a MUSE counterpart was found (no the extra sources, as we do not have mass information for them).*

  2. Boundaries in XY (proper-)distances were chosen to be as close as possible to the minimum and maximum size of the limits used to analyse the observations [$\sim (13 - 500)$kpc].
  3. Limit in the third dimension. Even though galaxies in the EAGLE simulation also have a z-position determined, this was not the case for our observations. The z-dimension for the last ones was estimated by the difference in its peculiar velocity (approximated by $v \sim cz$) as $\Delta v = |v_{p_{centre}} - v_{p_{neigh}}| < 2000 km/s$, in the case of spectroscopic redshifts. This was directly implemented

for the EAGLE counterparts, where the peculiar velocity-z of neighbours should not differ in more than 2000km/s with that one of the centre.

4. The snapshot number was selected to be the same as the centres, based on redshift proximity of galaxies within clusters.

For both cases, the selection of the Aperture Mass Radius was made following the recommendations of EAGLE documentation (=30pkcp). Although it is claimed that different apertures do not affect the simulations up to $M_* < 10^{11} M_\odot$, we also checked the results with an aperture of 10pckc as some of our centres were considerably close to this mass limit. As we can observe in Fig. 3.4, the average behaviour of sources effectively does not depend on the chosen aperture.

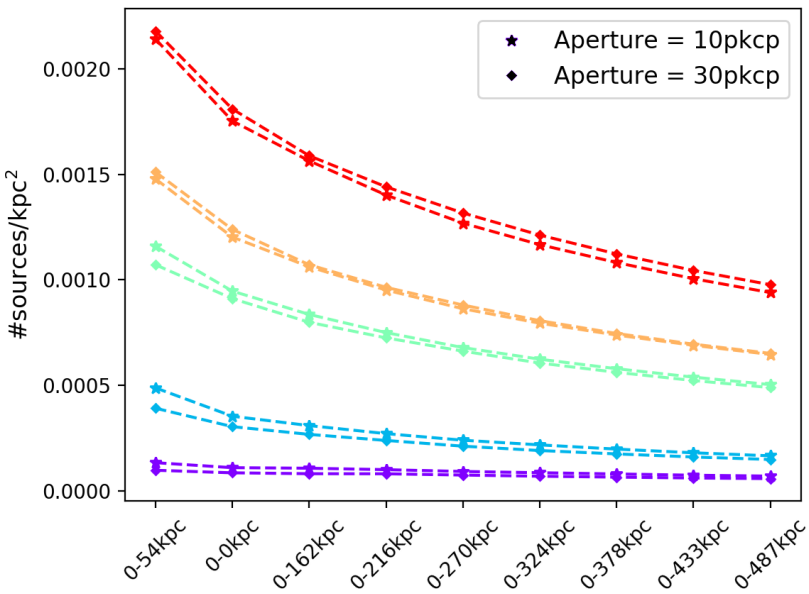


Figure 3.4: *Results obtained by using Aperture=10pkpc (stars) and Aperture=30pkpc (diamonds). Each colour correspond with a particular source, from up to down [ID_VLA: 4692, 8727, 4148, 3723 and 5063].*

3.2 MARZ: Automatic Redshifting Software

MARZ is the current redshift fitting software adopted for spectroscopic redshift determination for MUSE data[23]. Subsequently, this was the software used to determine the spectroscopic redshift of those sources only detected in the VLA catalogue, for which no MUSE counterpart was found (the 8 *extra sources*).

The project was first developed to take spectra form the AAOmega spectrograph to produce high quality spectroscopic redshift measurements. The resulting MARZ software[24] is a web-application that has been generalised to take spectra from any spectrograph, given a well defined .fits file input.

The central step of the automatic redshifting process lies in the matching procedure, where the observed spectrum is compared with a selection of 12 representative object type templates, simultaneously finding the best-fit object type and shifting it to the best-fit redshift.

Thanks to its interactive interface, a manual match option is also possible, allowing for simple verification of the top five automatic matches, all the way to assigning a determined object type to the spectrum and even manually entering a desired redshift.

Fig. 3.5 shows the initial overview screen where, after dropping the input .fits file in the designated box, users can, for example, choose the display data view, filter by object types, redshift range, etc. The second image correspond with the matching screen, where different menus allow the user to switch parameters on and off (Data, Template, Continuum...), see the top 5 automatic results, input a redshift manually, smooth the data, or select the template and type of object, among others. A confidence value is assigned via the vertical bottom menu on the left (Great: 3, Good: 2, Possible:1, Unknown: 0). And after finishing with the redshift determination, the results can be downloaded into a file containing information about all the analysis carried out.

To be able to use MARZ on our data we needed to first extract the MUSE spectrum of these sources, and later create a set of files readable by the software.

The first part of the process was to extract the spectra of these sources (making use of SExtractor[25]) from the MUSE datacubes for each CGR. We used a circular mask of radius $0.8''$ for the object extraction, and the COSMOS-HST image to which CGR belongs as the image from which cutouts are created. The extraction program will create a .fits file with the spectra of each extracted source (utilized to generate the ones we will drop

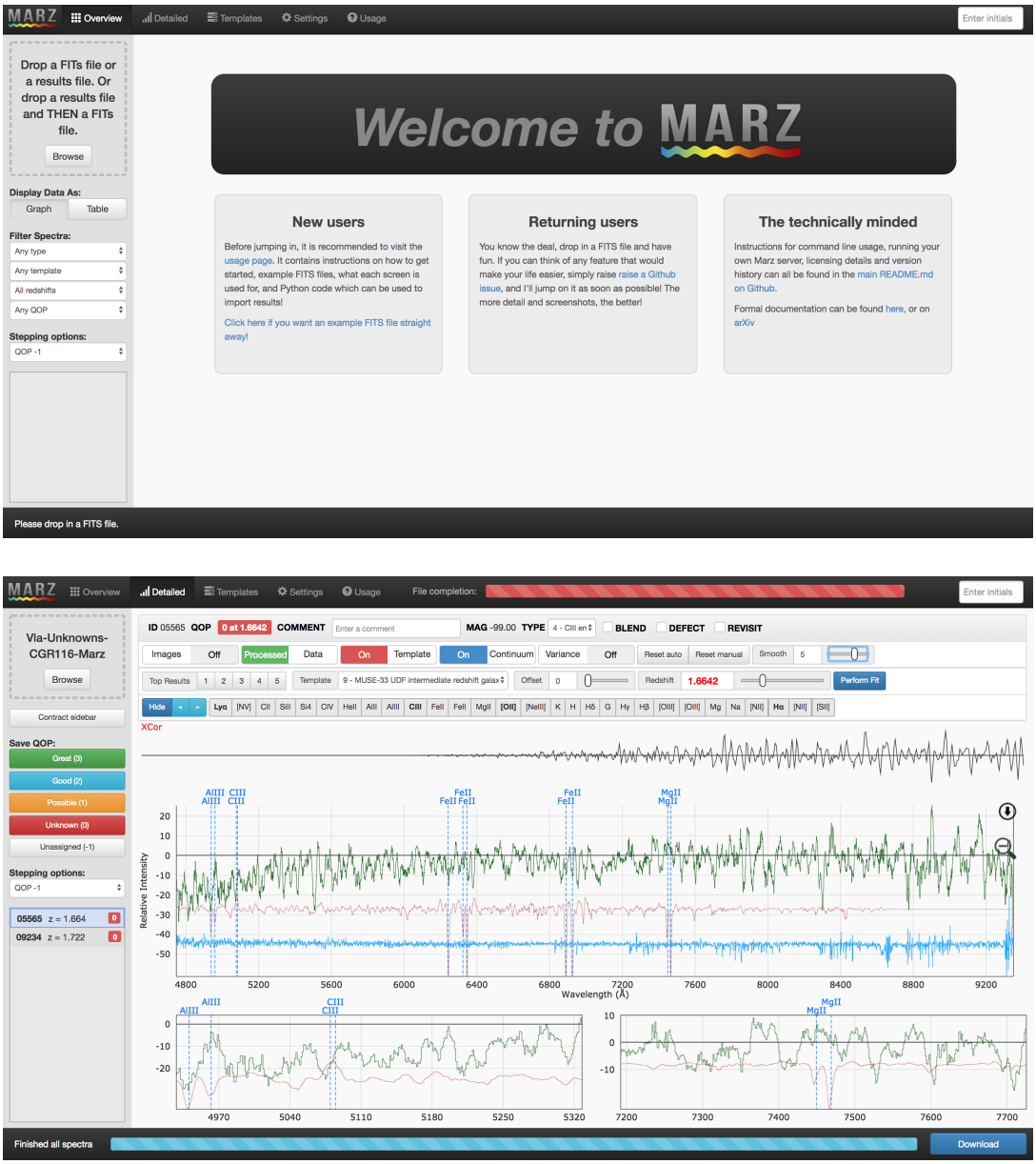


Figure 3.5: *The top image correspond to the initial overview screen, where we can navigate through the different content such as downloadable examples of input .fits files, users guide, etc. The dotted box on the top-left corner is the place designed to drop the input .fits files (we will call it input-box in the future). The data will be display in the main box, substituting the welcome message, and the view can be change through the options under the input-box. It also gives the option of filtering on object types, templates, redshift ranges, among others.*
The bottom image is the matching screen. The menus on the top allow to display or hide data such as template or continuum, smooth the data, or manually set a chosen redshift. It also allows to switch among the top five automatic results, and on the left, assigned a confidence value to the selected redshift. A download button in the bottom-right corner saves the results in a external file.

on the *input-box*) plus a .vot catalogue for each CGR, used to setup MARZ.

After starting MARZ software, it will create a server that we can open on a web browser, which will take us to the overview screen in Fig. 3.5. We can then drop the created .fits files in the *input-box* and start the redshift analysis.

Regarding the redshift analysis, once we launched the program, the first approach was to leave MARZ to do an automatic guess of the redshift. If the spectrum was considered not good enough in signal-to-noise ratio, the automatic redshift was assigned (unless the matched redshift was close to the photometric redshift, in which case the latest was assigned) with a confidence zero. Otherwise, we proceeded to manually study different possibilities and compare the results with the photometric redshift, a confidence was also assigned to each different chosen redshift.

3.3 Classification of sources

3.3.1 BPT diagrams

The galaxy activity has been categorised according to the relative intensity of different emission lines. More precisely, we adopted the widely used BPT diagrams[26,27,28], whose selected lines are based on the intensities of the collisionally-excited forbidden lines [OIII]λ5007, [NII]λ6584, and [SII]λ6717+6731 relative to that of the Balmer series recombination lines Hα and Hβ. A signal-to-noise ratio (S/N)≥3 was imposed for accurate classification of the galaxies into star-forming or AGN-dominated classes.

Our classification, largely following the criteria outlined in Kewley et al. 2006[27], comprises:

1. **Star-forming galaxies (SF):** classified as non-active in the [OIII]/Hβ–[NII]/Hα or [OIII]/Hβ–[SII]/Hα planes.

$$\log([OIII]/H\beta) < 0.61/[\log([NII]/H\alpha) - 0.05] + 1.3 \tag{3.4}$$

$$\log([OIII]/H\beta) < 0.72/[\log([SII]/H\alpha) - 0.32] + 1.30 \tag{3.5}$$

2. **Active Galactic Nuclei (AGN):** radiation powered by the accretion of matter onto a SMBH.

$$\log([OIII]/H\beta) > 0.61/[\log([NII]/H\alpha) - 0.47] + 1.19 \tag{3.6}$$

$$\log([OIII]/H\beta) > 0.72/[\log([SII]/H\alpha) - 0.32] + 1.30 \tag{3.7}$$

3. **Composite Galaxies (CG):** objects whose spectra may contain significant contributions from both a SMBH as well as star formation (they are likely to contain a young, metal-rich stellar population plus an AGN).

$$\log([OIII]/H\beta) > 0.61/[\log([NII]/H\alpha) - 0.05] + 1.3 \tag{3.8}$$

$$\log([OIII]/H\beta) < 0.61/[\log([NII]/H\alpha) - 0.47] + 1.19 \tag{3.9}$$

Fig. 3.6 shows where each galaxy activity is located in the diagnostic ratios [OIII]/Hβ with respect to [NII]/Hα (which is a good way of measuring the mean energy of an ionising photon) and [OIII]/Hβ with respect to [SII]/Hα.

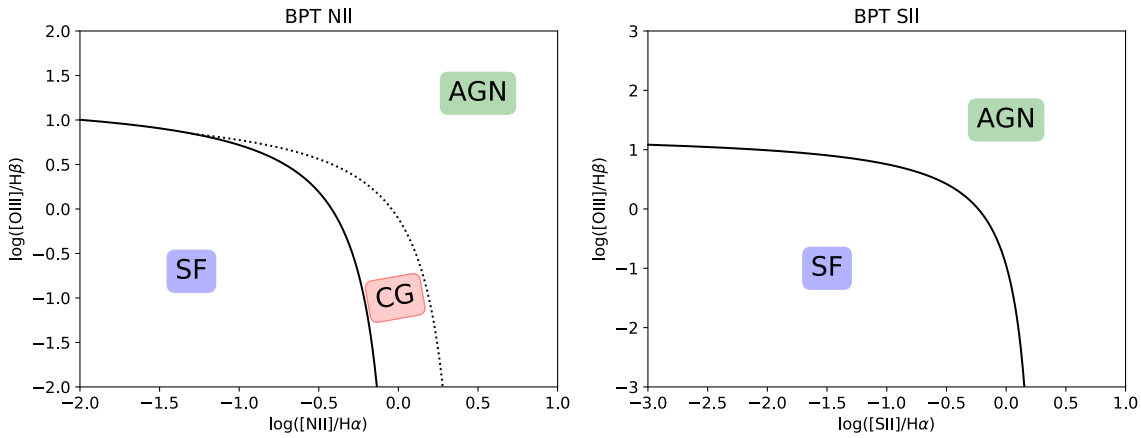


Figure 3.6: *BPT diagrams. Indicates where sources lay depending on its galaxy activity.*

3.3.2 Mass-Excitation diagrams

This classification method[29] is particularly useful for galaxies at intermediate redshift $0.3 \lesssim z \lesssim 1$ (not tested beyond $z \sim 1$), where classification through BPT diagrams becomes unavailable as the redder emission lines ([NII]λ6584/Hα) shift into the near infrared.

To solve this issue, Mass-Excitation diagrams adopt the stellar mass as a substitute of the aforementioned line ratio, justified by the suggested physical connection between mass and metallicity[30,31] ([NII]/Hα provides an indication of the gas-phase metallicity in SF galaxies[32]). Moreover, AGNs tend to be found in host galaxies with high stellar mass[28],

which puts them in a location on the modified diagram analogous to their original one on the BPT diagram. Fig. 3.7 shows the distribution of galaxy activity in this new classification method.

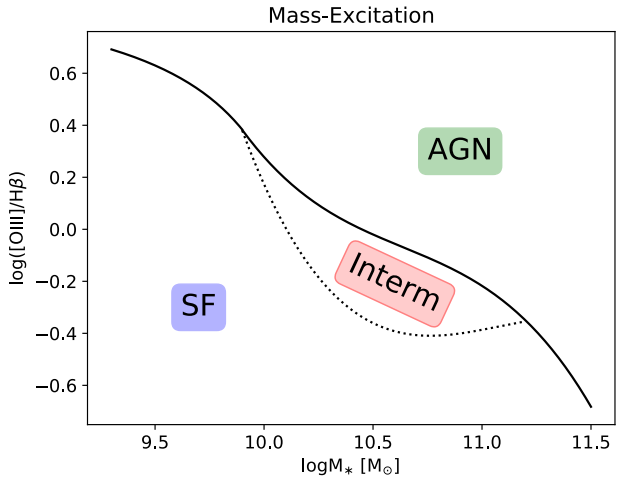


Figure 3.7: *Mass-Excitation diagram. Indicate where sources lay depending on its galaxy activity.*

Two empirical dividing lines are defined such that the separation between galaxy classes is maximised:

$$y = \begin{cases} 0.37/(x - 10.5) + 1, & \text{if } x \leq 9.9 \\ 594.753 + 167.074x + 15.6748x^2 + 0.491215x^3 & \text{otherwise} \end{cases} \tag{3.10}$$

$$y = 800.492 - 217.328x + 19.6431x^2 - 0.591349x^3 \quad \text{for } 9.9 < x < 11.2 \tag{3.11}$$

where $y = \log([OIII]/H\beta)$, and $x = \log(M_*)$

The galaxy activity is divided in:

1. **Star-forming galaxies (SF):** Placed below both lines.
2. **Active Galactic Nuclei (AGN):** Situated above the line defined in 3.10.
3. **Intermidiate (Interm):** Analogue to Composite Galaxies in BPT classification. Located in the region between the two lines.

3.3.3 'Lamareille et al.' diagrams

'Standard' diagnostic diagrams as BPT make used of 'red' emission lines, such as Hα, [NII]λ6584 or [SII]λ6717+6731, not observable in the optical range for redshift greater than $\sim$0.1.

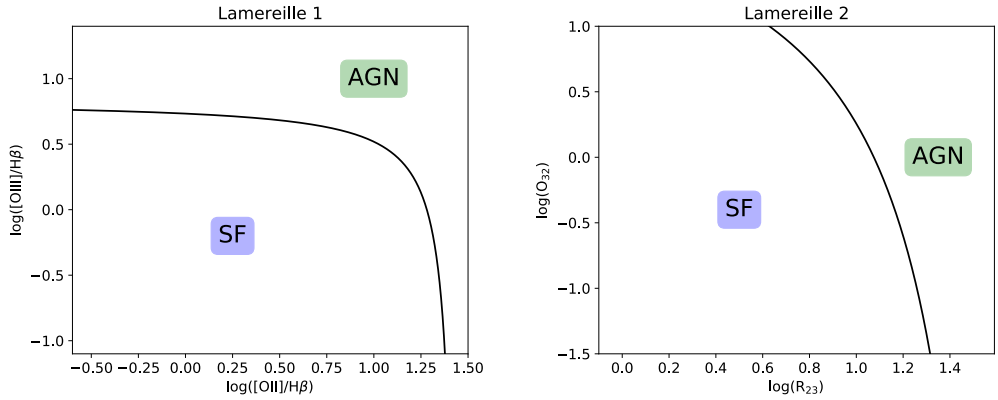


Figure 3.8: *Lamareille diagrams. Indicate where sources lay depending on its galaxy activity. The assigned title will be used along the paper when referring to one of the diagrams in particular.*

To efficiently separate AGNs from starbursts at intermediate redshift (z≥0.1), diagnostic diagrams must involve only 'blue' emission lines, i.e. [OII]λ3727, Hβ and [OIII]λ5007. We will follow the classification proposed by Lamareille et al. (2004)[33] (from now on mostly referred to as Lamareille diagrams), who define the following demarcation curve between starburst and AGNs:

$$\log([OIII]/H\beta) = 0.14/[\log([OII]/H\beta) - 1.45] + 0.83 \tag{3.12}$$

$$\log(O_{32}) = 1.5/[\log(R_{23}) - 1.7] + 2.4 \tag{3.13}$$ [¶]

As before, a S/N≥3 in the emission lines was imposed. Fig. 3.8 shows the classification of the galaxy activity within two groups:

1. **Star-forming galaxies (SF):** Placed at the bottom left in both diagrams.

2. **Active Galactic Nuclei (AGN):** Situated in the upper right of both diagrams.

3.3.4 VLA-COSMOS 3GHz source counterpart catalogue

Upon completion of the source type classification through the three aforementioned methods, 11 sources were still not categorised, 8 of them

[¶] $R_{23} = (O[III]\lambda 4959{+}5007 + O[II])/H\beta$ and $O_{32} = O[III]\lambda 4959{+}5007/O[II]$

being the VLA sources for which no MUSE counterpart was found. In addition, a total of 7 sources were only classified through one single method, making a fully justified selection of its galaxy activity impossible.

For this reason, we made use of the VLA-COSMOS 3GHz source counterpart catalogue[34], which classifies the sources as star-forming galaxies or AGN based on a multiwavelength criteria.

To complement the 3GHzRadio catalogue, they add information from the COSMOS2015[12] catalogue, which provides optical and NIR photometry, the *i* band selected COSMOS photometric catalogue[38], and the *Spitzer* - COSMOS IRAC catalogue[39]. In addition, they also use far-infrared photometry from the *Herschel* Space Observatory[40,41,42,43], and (sub)-millimetre photometry from the various observational campaigns over the field. X-ray band is covered by the COSMOS X-ray catalogue of point sources[44], sketched by the *Chandra* COSMOS-Legacy survey[45].

Cross-matching of sources among the different catalogues is mainly based on a nearest neighbour matching. In the case of the optical-MIR catalogues, each 3 GHz VLA source is associated with a single counterpart candidate following an ordered choice, when more than one match is available, from COSMOS2015, to *i* band and finally IRAC. With regards to the X-ray counterparts, the chosen catalogue has already a list with the IDs of COSMOS2015 and *i* band counterparts, which are directly assigned, while IRAC counterparts where cross-matched via a nearest neighbour matching.

Via a multiwavelength selection, the identified AGN/star-forming galaxy samples are categorised into three main classes, depending on the following criteria:

1. **Moderate-to-high radiative luminosity AGN (HLAGN):** These are sources that show signatures at wavelengths other than radio. Subsequently, they are based on a non-radio selection, as any radio excess in $\log(L_{1.4\,GHz}/SFR_{IR})$, i.e. the logarithm of the ratio of the 1.4 GHz rest-frame radio luminosity and SFR derived from the total IR emission in the host galaxy[46], is not taken into account. They can present an intrinsic [0.5-8] keV X-ray luminosity of $L_X = 10^{42}\mathrm{erg\,s^{-1}}$ (selected as X-ray AGN[47]), or be categorised as MIR-AGN via emission from a warm dusty torus, through IRAC colour-colour criteria[48]. They can also, even exclusively, present evidences of AGN activity on the basis of a SED-fitting analysis[49,50] (referred to as SED-AGN), that disentangles the possible AGN contribution from the host-galaxy light.

2. **Low-to-moderate radiative luminosity AGN (MLAGN):** They show AGN signatures other than X-ray, MIR or in the source SED.

   - **Quiescent MLAGN:** Following the UV/optical colour-based separation method developed by Ilbert et al. 2010[51], sources are classified as *quiescent* if $M_{NUV} - M_{r^+} > 3.5$ (red colour), where M_{NUV} represents the value of the rest-frame near ultraviolet, and being the r^+ band colours corrected for internal dust excitation. In addition, they need to not be detected in any of the *Herschel* bands. A large fraction of these AGN sources show also an excess of radio emission relative to the SFRs in their host galaxy.
   - **Radio excess MLAGN:** Objects with a $>3\sigma$ radio excess in the redshift-dependent distribution of $\log(L_{1.4\,GHz}/SFR_{IR})$. Part of the selected sources through this criteria overlap with the SFG sample (selected based on green and blue optical rest-frame colours), which is expected to have no radio excess, shedding light on the composite nature of the radio sources.

3. **Star-forming galaxies (SFGs):** Galaxies selected to have a dust-extinction corrected rest-frame colour $M_{NUV} - M_{r^+} < 3.5$ (define as *intermediate* and *high* activity galaxies), or $M_{NUV} - M_{r^+} > 3.5$ but also requiring to be detected in the *Herschel* bands. As it was mentioned before, part of this sample presents also a radio excess not proper of SFGs, enabling a composite character of the source. The sample obtained by excluding those radio excess sources from the SFG sample is define as the clean SFG sample, which most likely does not contain AGN host galaxies.

4. Results

4.1 Redshift

Determining the redshift of the sources in our sample, as well as its credibility, was a crucial component of our study, as it allowed us to define if a galaxy was considered a neighbour or not (part of the same galaxy cluster).

From the several available catalogues in the COSMOS field, we had access to both photometric and spectroscopic redshifts for a large number of sources. This allowed us to not only assigned a reliable redshift to most of our central sources, but also compare the accuracy of photometric with respect to spectroscopic redshifts for the MUSE sample.

In the case of the central galaxies, as was mentioned in Sec. 3.1.1, most of the radio-sources were matched with a MUSE counterpart (22 sources), and the MUSE catalogue employed had already computed spectroscopic redshift (using MARZ software) for all the observed sources. On the other hand, photometric redshifts were provided by the VLA-COSMOS 3GHz source counterpart catalogue[34], which contains what was considered as the most reliable redshifts available in the literature (some photometric redshifts are replaced by spectroscopic redshifts in this catalogue, when available, and the measurements are considered 'secure').

For those central radio-sources for which no MUSE counterpart was found (the remaining 8 galaxies), spectroscopic redshifts were found in a semi-automatic way using MARZ software, as described in Sec. 3.2.

Regarding the henceforth called neighbour sources, spectroscopic redshifts were again taken from the MUSE catalogue, while photometric redshifts correspond to those presented in the Laigle et al. catalogue[12], computed using LEPHARE[35,36,37].

Throughout the entirety of this study, it is important to mention that,

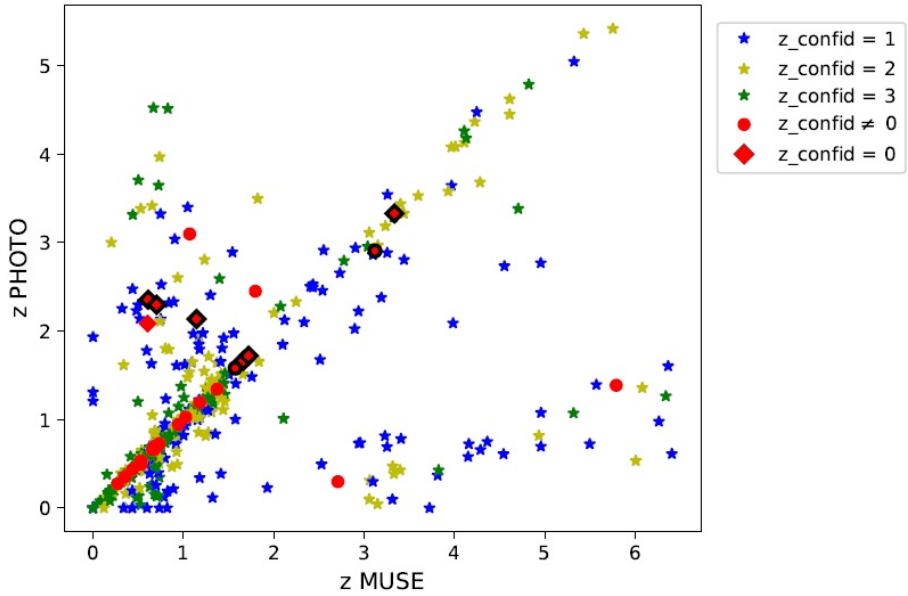


Figure 4.9: *Comparison of the spectroscopic (z MUSE) and photometric (z PHOTO) redshifts for all the sources in our analysis. Stars correspond with neighbour sources, whose spectroscopic redshift confidence from MARZ is equal to 1 (blue), 2 (yellow) or 3 (green). Red markers correspond with central galaxies of confidence 0 (rhombuses) or 1, 2, 3 (dots), and those one with a black edge are the sources for which no MUSE counterpart was found.*

except when explicitly stated otherwise, only spectroscopic redshift with positive confidence (different than zero, as this correspond to unknown redshift) are used. This is justified by the arguments followed to select the confidence for MARZ (spectroscopic) redshifts in the MUSE catalogue:

> **1:** `Possible redshift, determined by a single feature whose spectral identification remains uncertain.`
> **2:** `Secure redshift, determined by a single feature.`
> **3:** `Secure redshift, determined by multiple features.`

H. Inami et al., (2017)[‖]

[‖]Quotation from: H. Inami et al., *The MUSE Hubble Ultra Deep Field Survey. II. Spectroscopic redshifts and comparisons to color selections of high-redshift galaxies*, A&A **608** A2 (2017). **Page 4 of 26**

ID VLA	z MUSE	z confid	z PHOTO
3438	0.61003	0	2.3545
10586	0.70466	0	2.2968
4166	3.12045	1	2.906
7460	1.14556	0	2.1362
7409	5.79212	1	1.3863
8420	1.06903	1	3.0975
8727	2.71067	1	0.2973
7316	0.60532	0	2.084
9176	3.33674	0	3.3287
7700	1.57321	2	1.5811
8	1.79677	3	2.45
5565	1.6642	0	1.6642
9234	1.722	0	1.722

Table 4.1: *ID in the VLA catalogue, photometric and MUSE redshift, together with the assigned confidence to the MUSE redshift, for the remarkable sources.*

With the particular aim of debating the accuracy of our redshift estimates of the 8 sources lacking MUSE redshift (spectroscopic redshift in the MUSE catalogue), but extended to check the rest of the sample, we compare in Fig 4.9 photometric and spectroscopic redshift for all our sources, both central and neighbour galaxies. The sample is subdivided in terms of the confidence conferred to the MUSE redshifts, and specifying if they are centres or neighbours.

As we can see, both photometric and spectroscopic redshifts of our central radio-sources agree in most of the cases. The ones which do not agree but were already part of the MUSE catalogue, together with the *extra sources*, are summarised in Table 4.1, which contains the ID in the VLA catalogue, as well as photometric and MUSE redshift, with confidence.

In Fig. 4.9, the two central (red) sources above the line $zPHOTO = zMUSE$ with confidence $\neq 0$ (dots) correspond with sources already in the MUSE catalogue (subsequently not analysed in the framework of this project) [ID_VLA: 8420 and 8]. We also have another source from the MUSE catalogue over the equality line [ID_VLA: 7316], but with an assigned confidence $= 0$. The same happens with the two central sources under the line [ID_VLA: 8727 and 7490], but in this case both have confidence $= 1$, which as we said before, it is far from being trustworthy. In fact, ID_VLA: 7490 was not used in futures steps as not only the redshift was not trustworthy, but the assigned stellar-mass was excessively large

($\sim 10^{13} M_{\odot}$).

Regarding the redshifts of the *extra sources*, analysed during this research, they are described as follow:

- **CGR22 [ID_VLA: 3438]:** The spectroscopic redshift corresponds with the automatic one calculated by MARZ. The signal-to-noise ratio of the spectrum was not good enough to determine the redshift, and the confidence was assigned to be zero.

- **CGR22 [ID_VLA: 10586]:** As for the previous source, the spectroscopic redshift is taken from the MARZ automatic result. Almost no lines were clearly matched, and the spectrum was really noisy, thus the specified confidence is zero.

- **CGR24 [ID_VLA: 4166]:** This galaxy has a redshift confidence equal to 1. The automatic redshift from MARZ is in good agreement with the photometric redshift, and in addition, the spectrum present clear Lyα emission (some other lines relating to CIV, HeII are also visible, although the signal-to-noise ratio is much smaller in this case. See Fig 4.10). As the redshift was determined mainly based on the Lyα, with no more data to support the result, we decided to not assign a confidence higher than 1.

- **CGR24 [ID_VLA: 7460]:** The spectroscopic redshift of this source also correspond with that automatically calculated by MARZ. The signal-to-noise ratio of the spectrum was not good enough to determine the redshift, and the confidence was assigned to be zero.

- **CGR77 [ID_VLA: 9176]:** The analysis of the redshift of this source was a bit special. Although MARZ assigned an automatic redshift close to $\sim$1.5, the photometric redshift, assuming it is correct, would potentially allow us to see Lyα emission in the spectrum. We then decided to adjust the redshift close to the photometric one and look for the line, but the signal-to-noise ratio of the spectrum was not good enough to find any other line to support the result, and even the Lyα emission itself was not clear, thus the final confidence was chose to be zero.

- **CGR83_North [ID_VLA: 7700]:** The spectrum of this galaxy shows clear MgII and FeII absorption lines (see Fig. 4.11), and the photometric and spectroscopic redshift are in total agreement. The analysis of the spectroscopic redshift of this source was considered good enough to give a confidence of 2.

- **CGR116 [ID_VLA: 5565]:** In this case, the spectroscopic redshift was imposed manually to be the same as the photometric redshift, but no lines were detected and the confidence was assigned to be zero.
- **CGR116 [ID_VLA: 9234]:** As for the previous source, the spectroscopic redshift is taken to be the same as the photometric one. No lines were detected and the determined confidence is zero.

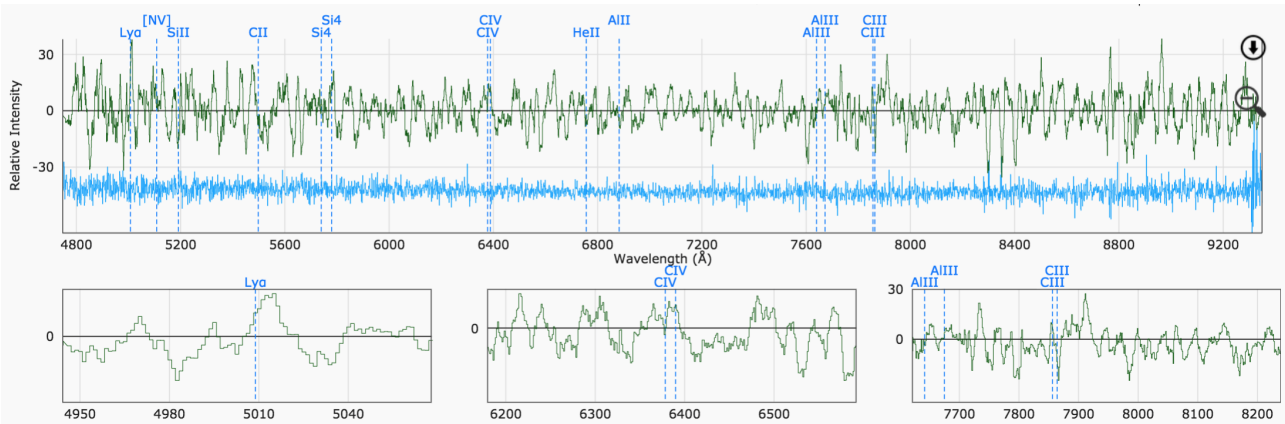


Figure 4.10: *Spectrum of the source ID_VLA: 4166 as seen in the MARZ matching screen. Continuum has been removed and some of the lines are enlarged on the bottom, e.g. Lyα emission can be seen in the first box.*

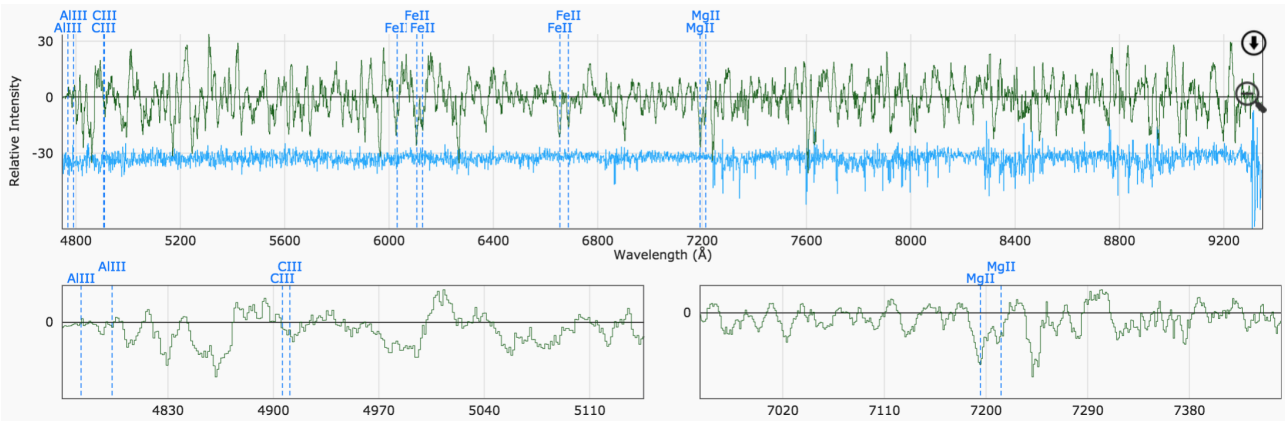


Figure 4.11: *Spectrum of the source ID_VLA: 7700 as seen in the MARZ matching screen. Continuum has been removed and some of the lines are enlarged on the bottom, e.g. MgII absorption lines can be seen in the box at the right.*

In general, the extracted MUSE spectrum of the *extra sources* was not good enough to provide a spectroscopic redshift confidence, and only 2

of them were determined with a confidence greater than zero [ID_VLA: 4166 and 7700]. Most of these sources have photometric redshift between $\sim$1.5$-$2.9, the so called *redshift desert*, which is a difficult range for MUSE spectra to allow redshift determination. Strong lines such as OII lie outside the wavelength range of MUSE at redshift higher than 1.5, and Lyα emission does not become detectable until redshifts around 2.9, these two being some of the most used tracers for redshift determination.

With respect to the rest of the sources in the MUSE catalogue (neighbours), the spectroscopic redshift and its confidence were already assigned, but we can see that for most of the galaxies with a redshift at confidence 3, photometric and spectroscopic redshift are in considerable agreement. The scattering is inversely proportional to the confidence, as expected. Sources with confidence 1, although being the least reliable, were still kept in our study.

4.2 Type of source

In order to study any type of possible correlation of the environment with the type of galaxy activity, we explored the nature of the radio sources, classifying them into SFG or AGN. With this purpose we made use of the different classifications available to our sample, described in Sec. 3.3.

Table 4.2 shows the classification of each source in each of the diagrams used, plus the classification from the VLA-COSMOS 3GHz source counterpart catalogue. The last column corresponds to the selected galaxy activity after analysing all the different results.

Due to the fact that most of our central sources have redshifts higher than the 0.1 limit of effectiveness for BPT diagrams, only 2 sources could be classified by this method. On the other hand, the Mass-Excitation diagram allowed to classify a large number of sources (14), as the limitation on the redshift is reduced with respect to BPT diagrams. Lastly, the Lamareille diagrams provided information of up to 17 sources, that were classified in at least one of the two diagrams.

Regarding the Smolčić column, the type of source has been selected as followed:

- **SFG:** If categorised as CleanSFG in the catalogue.
- **XrayAGN/SEDAGN/MIRAGN:** If it was identified as HLAGN only through one of these criteria.

ID VLA	BPT NII	BPT SII	Lamareille 1	Lamareille 2	Mass-Excitation	Smolčić	Source Type
5063	SF	SF	SF	SF	SF	SFG	SFG
5720	not classified	not classified	SF	SF	interm	SFG	SFG
3723	not classified	not classified	not classified	not classified	AGN	MLAGN	AGN
5733	not classified	not classified	not classified	SF	not classified	SEDAGN	unclassified
6950	not classified	not classified	not classified	SF	not classified	SFG	SFG
4860	SF	SF	SF	AGN	SF	SFG	SFG
9539	not classified	not classified	SF	SF	SF	SFG	SFG
7409	not classified	not classified	not classified	not classified	not classified	SFG	SFG
4148	not classified	not classified	SF	SF	interm	SFG	SFG
8420	not classified	not classified	not classified	SF	not classified	SFG	SFG
1017	not classified	not classified	SF	SF	AGN	MLAGN	unclassified
8727	not classified	not classified	not classified	not classified	not classified	MIRAGN	AGN
9515	not classified	not classified	not classified	SF	not classified	SFG	SFG
7316	not classified	not classified	not classified	not classified	AGN	XrayAGN	AGN
2454	not classified	not classified	SF	SF	SF	SFG	SFG
2022	not classified	not classified	SF	AGN	AGN	XrayAGN	AGN
4692	not classified	not classified	SF	SF	AGN	XrayAGN	unclassified
8820	not classified	not classified	SF	SF	AGN	HLAGN	unclassified
2563	not classified	not classified	SF	SF	interm	SFG	SFG
8741	not classified	not classified	not classified	SF	not classified	SEDAGN	unclassified
8	not classified	not classified	not classified	not classified	not classified	HLAGN	AGN
6219	not classified	not classified	SF	SF	SF	MLAGN	unclassified
3438	not classified	not classified	not classified	not classified	not classified	SFG	SFG
10586	not classified	not classified	not classified	not classified	not classified	SEDAGN	AGN
4166	not classified	not classified	not classified	not classified	not classified	SEDAGN	AGN
7460	not classified	not classified	not classified	not classified	not classified	SFG	SFG
9176	not classified	not classified	not classified	not classified	not classified	SFG	SFG
7700	not classified	not classified	not classified	not classified	not classified	SFG	SFG
5565	not classified	not classified	not classified	not classified	not classified	composite	unclassified
9234	not classified	not classified	not classified	not classified	not classified	XrayAGN	AGN

Table 4.2: *Contains ID in the VLA catalogue, galaxy activity using the five different diagrams described in Sec. 3.3 (each column correspond with the name given to the respective figure of the diagram), the one provided by the VLA-COSMOS 3GHz source counterpart catalogue*[34]*, as well as the final decision after comparing all the results.*

- **HLAGN:** If it is the result of a combination of two or more criteria among X ray, SED-fitting or MIR colour-colour.
- **MLAGN:** If it is a quiescent galaxy with radio excess.
- **Composite:** If it is classified as both SFG and AGN in the catalogue.

Our selection was made prioritising the results from the counterpart catalogue, as the multiwavelength criteria provide a more trustworthy result than pure optical analysis. This allowed us, for example, to filter those sources that had one diagram in disagreement with the rest. For those cases when it was the only classification available, the type of source was directly selected from this catalogue. With regards to the unclassified sources, they correspond to what we considered to be possible composite galaxies, since they mostly classify as SF in our used diagrams but are classified as AGN in the counterpart catalogue.

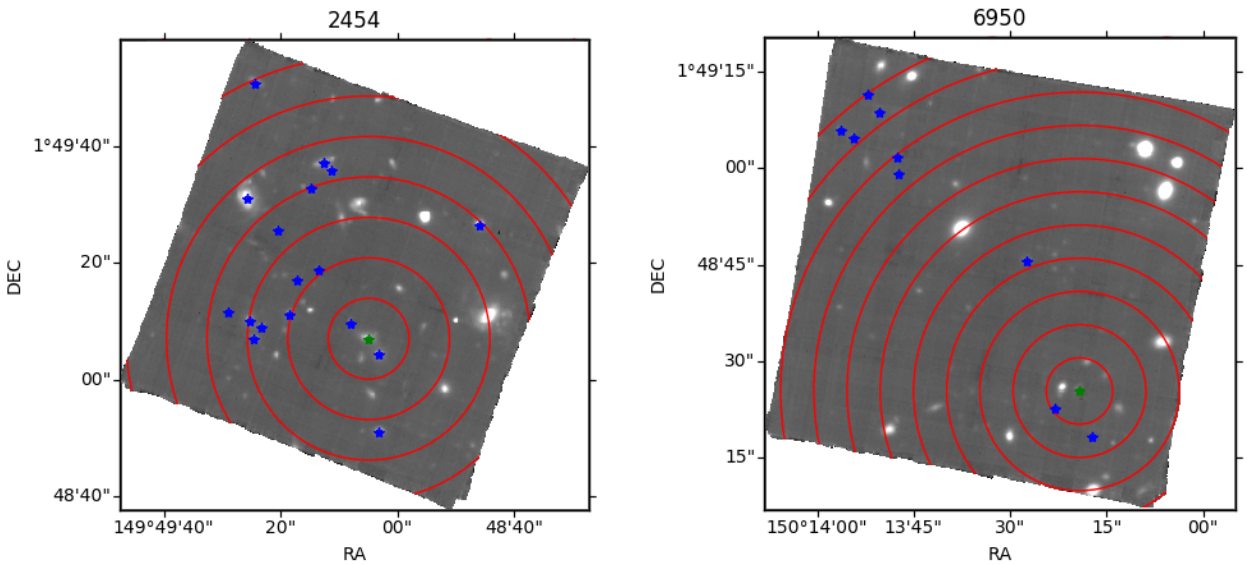
In order to compare the behaviour of radio detected sources with those only present in the MUSE catalogue, the latter were classified using both Lamareille and Mass-Excitation diagrams. The study was focused on AGN galaxies, classified as such only if its spectroscopic redshift had a confidence different than zero, and were classified as AGN in at least 2 out of the 3 diagrams. A total of 10 galaxies throughout the 10 COSMOS-groups used along this project were classified as AGN by the aforementioned condition.

4.3 Environment of radio sources

The main goal of the project lied in the analysis of the environment of the selected radio sources. To study the evolution of the density distribution of surrounding galaxies, in terms of the distance to the centre, we subdivide our field of view in rings for which we will count the number of sources per area (see Fig. 4.12).

As we can see in the immediately aforementioned figure, due to the position in the MUSE FoV of our radio sources, most of the time this area was not the totality of the ring, but a part of it. This made our counting to be limited by the available area, ignoring everything that is placed outside, and as such we do not limit ourselves to counting galaxies but normalise them to the studied ring section, thereby 'accounting' for the non-seen sources.

It is important to mention that the majority of the study is focused on sources whose spectroscopic redshift confidence is greater than zero, al-



(a) *CGR77−ID_VLA: 2454* ***(b)*** *CGR26−ID_VLA: 6950*

Figure 4.12: *Represent the different rings in which we subdivide the FoV. The green star represent the position of the central radio-source, while blue are the galaxy neighbours as explain in Sec. 4.3.1.*

though photometric redshifts and zero-confidence sources have also been explored.

4.3.1 Definition of neighbours

The first thing to do is to decide what we will consider by environment (limits on our galaxy neighbours). For this, it was necessary to define both distances in the XY plane (rings), plus a third dimensional distance (in redshift).

Due to the limitation by the MUSE FoV in XY plane, the radius of rings was selected to be up to the maximum distance where it was possible to detect a galaxy, in any of the CGR. This means that, to keep a consistent definition, all sources have the same ring-size calculated by a subdivision of 10 numbers in equidistant intervals between 0kpc and r_{max}, where r_{max} corresponds with the proper-distance in XY of the farthest source we were able to detect for one (not determined) of the central sources. This results in a total of 9 rings for which we will calculate the number of sources that lie within.

Regarding the Z-dimension, it was calculated in the basis of a redshift

relation (difference between peculiar velocities). As was mentioned while discussing the definition of neighbours in the EAGLE sample, this third dimension was restricted by a maximum velocity difference between two galaxies to be considered in the same large galaxy cluster, more specifically $\Delta v = c|z_{centre} - z_{neigh}| < 2000 km/s$ when redshifts were spectroscopic, and $\Delta v = c|z_{centre} - z_{neigh}| < 10000 km/s$ when redshifts were photometric. This divergence resides in the errors considered for each type of redshift, while the redshift/velocity uncertainty for spectroscopic redshifts from MUSE analysis have been computed to be on the order of[23] σ_z =0.0001(1+z), photometric errors are[12] $\sigma_z \sim 0.009(1+z)$.

Once we have the environment volume define, we can proceed to calculate the density distribution of galaxies around the central radio-sources.

4.3.2 Evolution of the density distribution

First of all, we consider the distribution of galaxies for which spectroscopic redshifts with confidence greater than zero exist. This results in a total of 23 central radio-sources for which Fig. 4.13 shows a behaviour example.

As we have mentioned several times before, not all the galaxies in the MUSE CGR catalogues have spectroscopic redshifts confidences greater than zero. To account for those 'missed' galaxies we define a scale factor as:

$$f_{scale} = \left.\frac{\text{\# photo-galaxies}}{\text{\# muse-galaxies}}\right|_{\text{in total area}} \tag{4.14}$$

where # *photo-galaxies* is the number of neighbours using photometric redshift (which should account for all the galaxies fulfilling the neighbour criteria in the FoV), and # *muse-galaxies* corresponds to the number of galaxy neighbours for which spectroscopic redshift with confidence greater than zero is available. Both correspond to the total number counted within the total area (a sector-circle within the FoV of radius r_{max} centred at the radio-source).

We said that the the number of neighbours using photometric redshift should account for all the galaxy neighbours, but we encountered that this was not always the case. Since the same number of neighbours using photometric redshift should be at least the same as those using spectroscopic redshifts, the cases where

Since there should be at least the same number of neighbours using photometric redshift than those using spectroscopic redshifts, the cases where

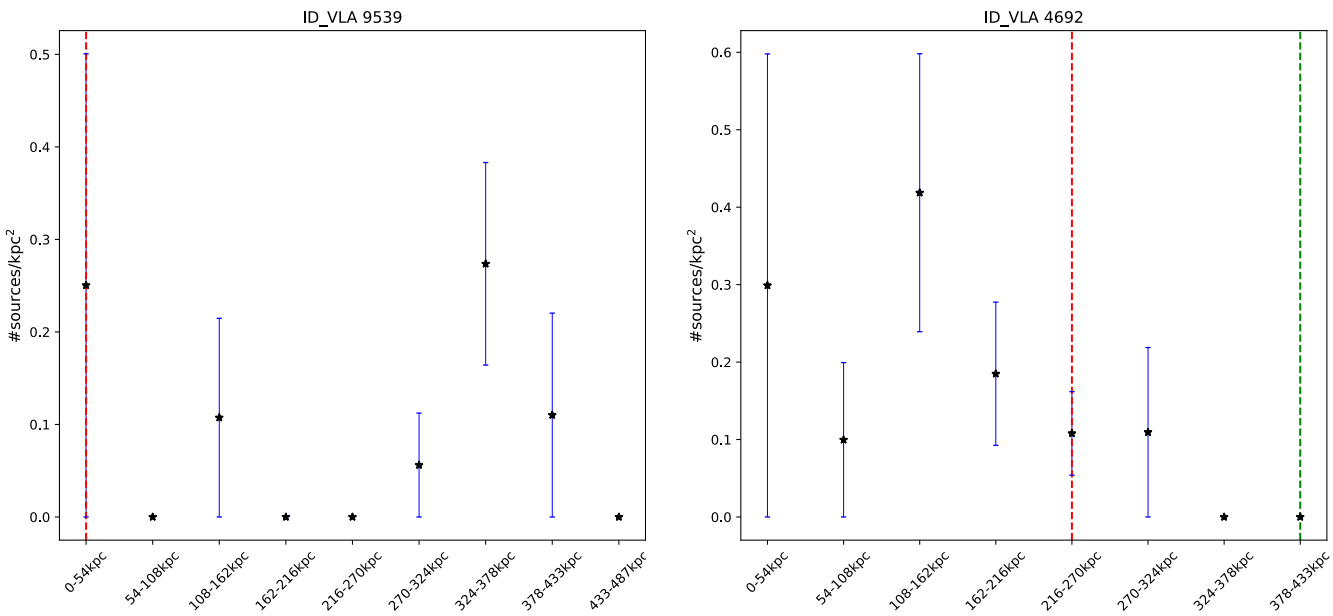


Figure 4.13: *Evolution of the density distribution of sources [ID_VLA: 9539 and 4692]. The dotted-red line represent the first bin where part of the ring area lies outside the FoV, while the dotted-green line correspond with the moment when the totality of the ring area is outside the FoV (this stop the analysis as there are no more sources to count and the area to divide is zero).*

$$\text{\# photo-galaxies} < \text{\# muse-galaxies} \longrightarrow f_{scale} = 1 \tag{4.15}$$

Then, the total number of sources per ring-area will be:

$$\begin{matrix}\text{\# sources per} \\ \text{ring-area}\end{matrix} = \begin{matrix}\text{\# muse-galaxies} \\ \text{per ring-area}\end{matrix} \times f_{scale} \tag{4.16}$$

where f_{scale} takes the values of eq. 4.14 and 4.15.

We also have some galaxies that do not present neighbours when the used redshifts are spectroscopic (# muse-galaxies per ring-area $= 0$), but do when using the photometric. For those cases, the number of sources per area would correspond directly with the results found through the

photometric analysis.

$$\begin{matrix}\text{\# sources per} \\ \text{ring-area}\end{matrix} = \begin{matrix}\text{\# photo-galaxies} \\ \text{per ring-area}\end{matrix} \tag{4.17}$$

Table 4.3 shows the # muse-galaxies (MUSE) and # photo-galaxies (PHOTO) for each source with # muse-galaxies per ring-area $\neq 0$, as well as the assigned f_{scale}.

ID_VLA	MUSE	PHOTO	f_{scale}
1017	21.0	12.0	1
2022	16.0	15.0	1
2454	17.0	14.0	1
2563	25.0	36.0	1.44
3723	13.0	11.0	1
4148	43.0	22.0	1
4692	16.0	15.0	1
5063	8.0	9.0	1.125
5720	1.0	10.0	10.0

ID_VLA	MUSE	PHOTO	f_{scale}
5733	1.0	3.0	3.0
6219	11.0	6.0	1
6950	9.0	9.0	1.0
7700	1.0	2.0	2.0
8741	6.0	6.0	1.0
8820	8.0	13.0	1.625
9515	2.0	2.0	1.0
9539	9.0	7.0	1

Table 4.3: *Values of # muse-galaxies (MUSE) and # photo-galaxies (PHOTO) for each source with # muse-galaxies per ring-area $\neq 0$, as well as the assigned f_{scale} according to eq. 4.14 and 4.15.*

On the other hand, the behaviour of the density distribution seems to be closely dependant of the ring-size, as we can appreciate in the noisy curves in Fig 4.13, most likely because of the fact that our areas are not perfect rings, but sectors of a ring, and we are not counting all the galaxies around the source. If for example, we encounter an overdensity in our field of view and the area is a ring-sector (as is the case of the last ring of Fig. 4.12b), our method will extrapolate that overdensity to the total ring-area, increasing unrealistically the number of neighbours. This problem was solved by means of a cumulative distribution, where the number of sources in each bin is added to the next bin, and also substituting the ring-area by a circle-area (cumulative ring). In this way, we will count in each bin the total number of sources up to that limit, normalised to the total area covered. This approach helped to dilute the overdensities and decrease the noise in each curve (see Fig. 4.14 for a comparison).

The errors in the number of sources per area, in absence of different observations or models, were calculated through bootstrapping. The tech-

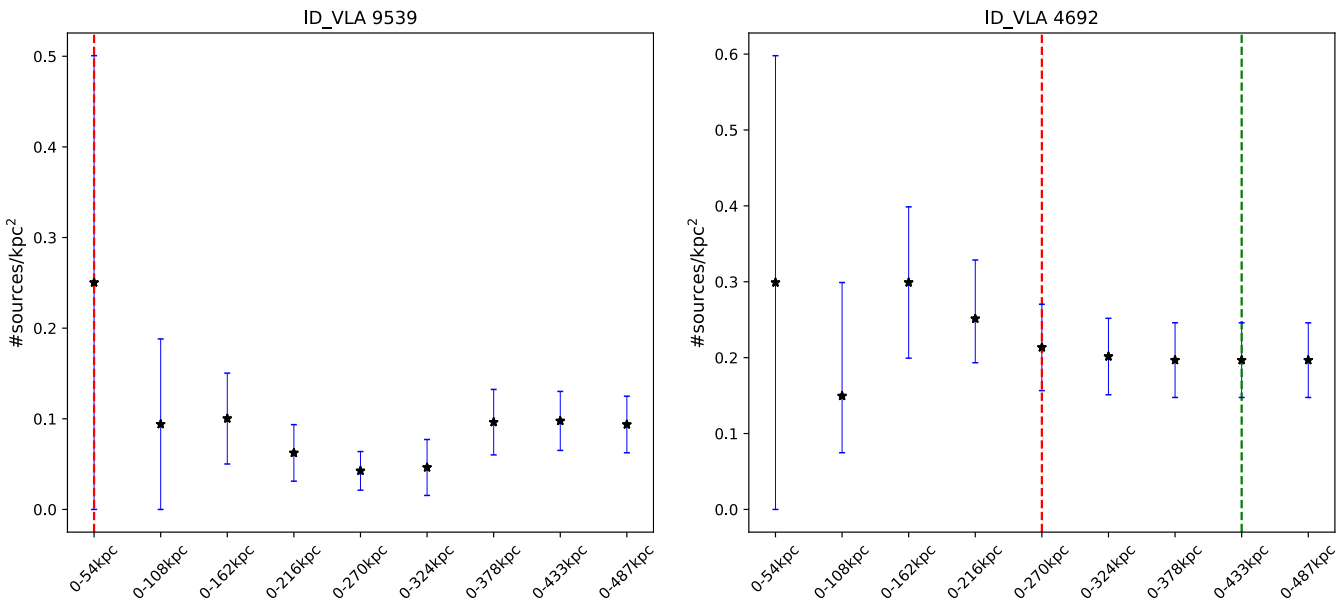


Figure 4.14: *Evolution of the cumulative density distribution of sources [ID_VLA: 9539 and 4692]. The dotted-red line represent the first bin where part of the ring area lies outside the FoV, while the dotted-green line correspond with the moment when the totality of the ring area is outside the FoV. Notice that, as the areas are now circles, the analysis does not stop after the dotted-green line. The surface after this line now correspond to the totality of the field.*

nique was implemented by generating 1000 new catalogues for each CGR based on the original one, randomly selecting galaxies from the MUSE catalogue, with replacement (same galaxy can be selected more than once), until ending up with the same number of total sources as the original catalogue. Then we run the same program as for the model on the new data, obtaining their respective amount of sources per area, and calculating the upper and lower error as the 84 and 16 percentiles respectively. It is important to mention that, as we are representing cumulative distributions, the errors are also cumulative (thus they are that large in the tail). Also note that the error for the initial ring is highly influenced by how we implemented the bootstrapping technique, which usually did not take any of the sources, in itself not really numerous, within this small area.

4.3.3 Comparison with EAGLE

In order to test our results, we made use of the EAGLE hydrodynamical simulation (refer to Sec. 3.1.2 for a full description of sources selection and properties). This comparison also allowed us to study the possible EAGLE limitations on this type of research.

The first step was to compare our results with those of the EAGLE counterpart without taking into account the source-type, i.e. the behaviour of the 100 counterpart centres was averaged and directly compared with the observational results. In order to compare the trend of the curves, they were normalised to their value at $r = 150kpc$. Fig. 4.15 shows some examples, where, as before, both errors and curves are cumulative.

We observe that for most of the sources, EAGLE results are able to reproduce the trend of the observations. But as we also noticed in Fig. 3.4 while comparing the influence of the aperture size, its evolution of the density distribution is almost the same for all sources. This appreciation is reenforced by the representation of both the EAGLE and MUSE slope curves in function of the stellar-mass of the central source, by a linear fitting $\log_{10} y = \alpha \log_{10} x$ (see Fig. 4.16). While MUSE values are mostly scattered, EAGLE slopes all lie in a straight line around -0.3, which confirms that effectively the average behaviour is the same, and proving also that the EAGLE simulation is complete in a statistical sense.

To go a bit further in our testing, since we also had information about the source-type, for both our observed galaxies and the EAGLE counterpart, we decided to explore next if the matching of the evolution of the density distribution could improve by also restricting the activity of the EAGLE centres. We then averaged the behaviour of those galaxies among the 100 counterpart centres that have the same activity as the observed radio-source, and as before, normalised both curves to their value at $r = 150kpc$. These results are shown in Fig. 4.17 for the same centres as Fig. 4.15.

We can see that, in general, restricting the source-type seems to improve in particular the initial behaviour [ID_VLA: 5063, 6950 and 8820], close to the source. But also there are some cases where the slope totally changes [ID_VLA: 3723], giving 'worse' results.

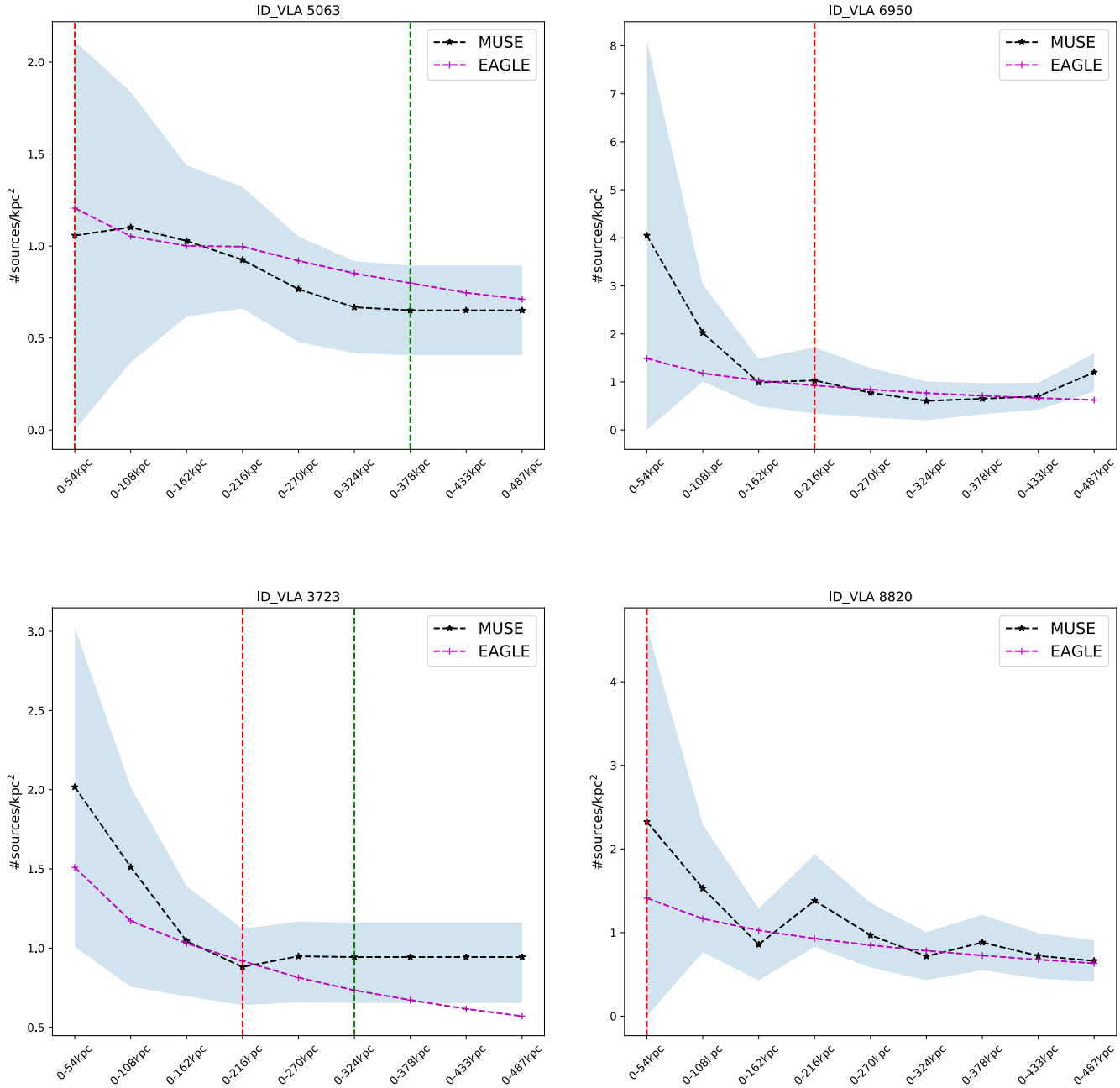


Figure 4.15: *Evolution of the cumulative density distribution for both MUSE and EAGLE data, without restriction in source-type. The dotted-red line represent the first bin where part of the ring area lies outside the FoV, while the dotted-green line correspond with the moment when the totality of the ring area is outside the FoV.*

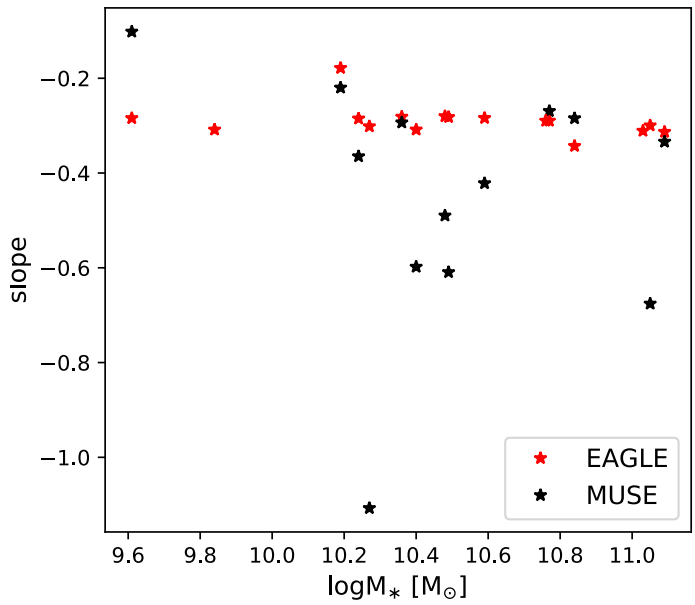


Figure 4.16: *Slopes of the MUSE (black markers) and EAGLE (red markers) cumulative curves, with no restriction in the activity of the central galaxy. Important to notice the number difference in MUSE and EAGLE sources, due to the fact that some centres had no observed neighbours, while they present some in the simulation*

In order to study a possible correlation of the initial slope with the type of source, we represent in Fig. 4.18 the slopes of both MUSE and EAGLE curves, as before, by a linear fitting $\log_{10} y = \alpha \log_{10} x$, with respect to the stellar-mass of the central source. As we can see, the slopes scatter around the same area, which fits with what we can observe in the comparison plots (in general, the observations and EAGLE simulation behaves really similar). But, on the other hand, they are not grouped by type of source or other property that make us think about a direct dependence. Also note that now the EAGLE curves are the result of a much smaller average sample, as we are narrowing down by source-type, which explains the scattering in the slopes with respect to Fig 4.16.

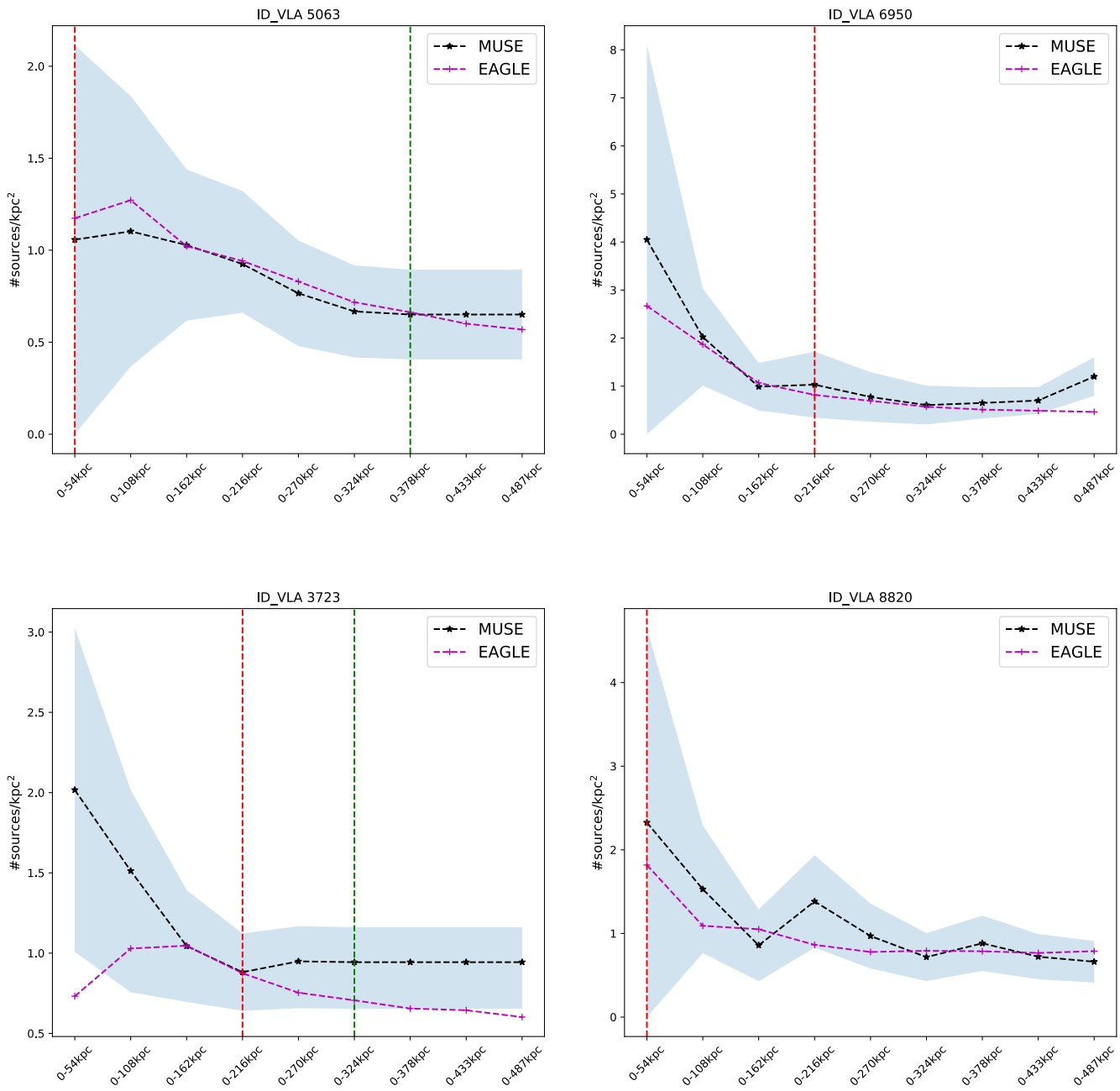


Figure 4.17: *Evolution of the cumulative density distribution for both MUSE and EAGLE data, with restriction in source-type. The dotted-red line represent the first bin where part of the ring area lies outside the FoV, while the dotted-green line correspond with the moment when the totality of the ring area is outside the FoV.*

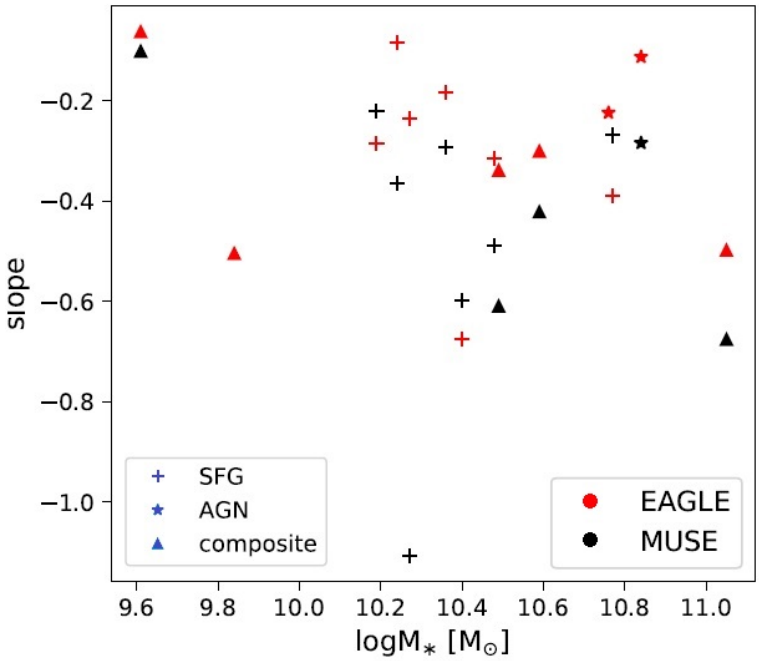


Figure 4.18: *Slopes of the MUSE (black markers) and EAGLE (red markers) cumulative curves, restricting the activity of the central galaxy. The source-type correspond with SFG (crosses), AGN (stars) and composite (triangles). Important to notice the number difference in MUSE and EAGLE sources, as well as in source-type. This is due to the fact that some centres had no observed neighbours, while they present some in the simulation, and that was not always possible to find a match in type of source among the 100 EAGLE counterparts.*

5. Discusion

5.1 Properties of the sources

The existence of the MUSE-GTO data, with a total number of 1662 identified sources, allowed us to derive spectroscopic redshifts for $\sim$ 55% of the sample. Photometric redshifts are known for 75% of the sources, which means that most of the galaxies within the catalogue, with a determined redshift, have an associated spectroscopic redshift.

A good spectroscopic redshift determination is crucial to reduce the scale of the galaxy environment study. It allows for, among others, a better determination of the Z-axis, as spectroscopic redshifts are usually estimated with a much smaller error than the photometric ones.

The attempt of determining spectroscopic redshifts for those 8 *extra sources*, that initially were not identified in the MUSE sample, had a successful result in 2 of the cases. The main reason of the failures resides in the fact that the MUSE spectra had a too low signal-to-noise ratio for a good matching of lines. We mentioned that these sources had an assigned photometric redshift within the range $1.5 - 2.9$, difficult to analyse with MUSE data. But also, if we have a look at the successful rate of spectroscopic redshift determination as a function of exposure time, 75% of the sources in CGR28 (observed during 9.75h) possess spectroscopic redshifts with confidence $\neq 0$, and 62% for CGR83, with 5.25h. This shows that the quality of the spectra used for determining redshifts increments exponentially with the exposure time, and that more hours of observation would potentially lead to a better analysis.

Regarding the classification of the central sources with respect to its galaxy activity, the performance of a multiwavelength analysis, together with the used of several diagrams in the optical range, allowed us to assure a better justified source-type selection.

The other side of performing more than one analysis method, although not necessarily a negative, is the variety in the results that each approach can obtain. The main nature of these disagreements between classification diagrams is due to the fact that galaxy activity is being analysed in its entirety. We do not have enough resolution to be able to separate nuclear activity from disk activity efficiently, and star-forming regions are mixed with AGN activity. Subsequently, our classification is based on the most prominent signatures, i.e. a source being classified as AGN does not mean that it does not have any star-forming region, only that the nuclear activity tracer is more notable. This is the reason why some diagrams (due to the different lines used on the classification) can differ on the source activity depending on the importance of AGN or SF signatures. The multiwavelength criteria helps to account for the most common classification, and distinguish better between contaminated tracers and truly classified source-type.

5.2 Neighbour determination

The number of sources per area with assigned spectroscopic redshift was high enough to perform a study of the density of galaxies around radio-sources by an individual counting. This also allowed us to consider regions really close to the centre, opening a new research on smaller scales.

The first problem encountered on this analysis links with how scattered the different MUSE COSMOS-Groups are, together with the relative small FoV with respect to the total COSMOS field. If the position of the radio-source was close to the edge of the observed region, the number of sources per available area was totally biased by the existing overdensities. The problem was solved by increasing the analysed area by substituting the ring-areas by circle-areas, and counting the total number of sources within them (cumulative sample). This analysis statistically dilutes overdensities in determined small areas, by performing an 'average' between overpopulated and isolated regions.

In future researches, the ideal scenario would be to find an equilibrium between field coverage of a bigger region, and high quality resolution that allows spectroscopic redshift determination.

Once we solved the issue with the XY-plane, the next step in the definition of neighbour galaxies lied in the determination of a Z-distance. When spectroscopic redshifts are well known for both centre and surrounding galaxies, our approach is able to successfully find all the observed neighbours within the FoV. But in order to account for those 'photometric neigh-

bours', we used a scale factor (Eq. 4.16) that we mentioned should ideally be always greater than one, although this is not the case for all the centres. Sometimes the number of neighbours by means of photometric redshift is smaller than the one found using spectroscopic redshift, mostly due to the error in the photometric redshift determination, and the main reason why this type of research needs high quality spectroscopic redshifts, as was mentioned previously.

We tried to account for this photometric error by setting the limit in the maximum velocity difference to $\Delta v < 10000 km/s$ instead of $2000 km/s$ as for spectroscopic redshifts, but the obtained result is that this change is not enough. Where should we put the limit then? Increasing the velocity difference would also result in the introduction of spurious results, so continuing with this same approach will not be a solution. For future studies, the method of calculating the number of neighbours when using photometric redshift should be different than a one-by-one counting with a modified Z-distance, or ideally all redshifts used should be spectroscopic.

5.3 Density distribution dependence

We computed the evolution of the density distribution of galaxies with respect to the projected distance to each central radio-source. In general, with a couple of exceptions, higher density regions are located close to the source in the XY-plane, decreasing logarithmically with distance. This is also reproduced by the EAGLE simulation, that obtains really similar results, especially when specifying the source-type.

We can confirm that EAGLE has successfully passed the first steps in testing how well it behaves at these scales in reproducing observations, and that, based on this research, it could potentially be used in the future for studies of close galaxy environments in compliment with observations. Although, as we will conclude later, our observed sample is not big enough to make certain conclusions, the chances of studying only sources that behave as EAGLE, and that increasing the number of targets will distance from the simulation, are really small.

In addition, we have information on stellar-mass, source-type and redshift for most of the radio-centres (excluding the 8 *extra sources*), for which we would like to explore if there is any dependence between the distribution of its galaxy neighbours and any of the centre's properties.

Fig. 5.19 shows the total number of neighbours within a circular area (restricted to the FoV) of radius $r_{max} \approx 487 kpc$, for each central radio-source, in function of redshift, stellar-mass and source-type. The number

of sources per area are scaled as indicated by Eq. 4.16 and 4.17.

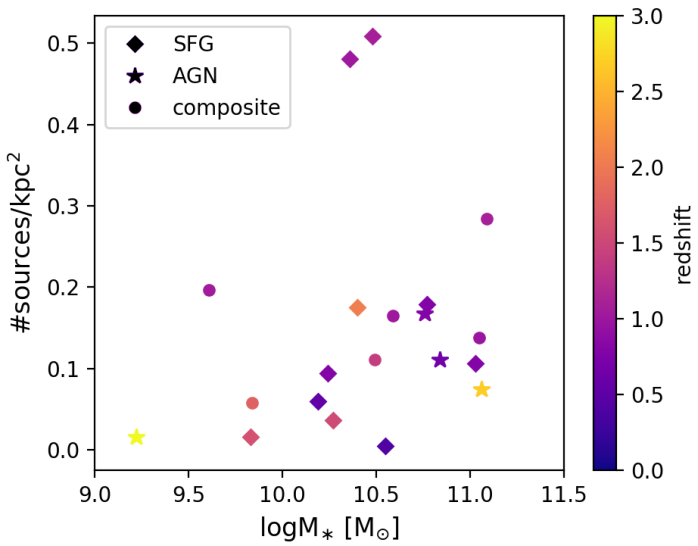


Figure 5.19: *Total number of neighbours per area (circle of radius $r_{max} \approx 487kpc$, restricted to the FoV) as a function of stellar-mass of the central source. Results are compute using the observational data. It also shows the redshift of each source as well as its galaxy activity.*

Before starting a global analysis, we need to think of how each variable individually affects the number of neighbours.

We expect to have more neighbours at lower redshifts, for a simple limit in the observations. Our capability of observing galaxies decrease with redshift, as it exists a limit in the luminosity for each redshift up to which we are no longer able to detect any source.

Assuming from the relative success of the matching of the EAGLE simulation to our observations that both will behave in the same way, and from the fact that it is not restricted by observational limits (no dependence on redshift), we can estimate the dependence of the total number of neighbours per area with the stellar-mass of the centre. Fig 5.20, shows no clear relation between stellar-mass and density-environment, which contradicts the results observed at larger scales, where the characteristic stellar mass of galaxies increase as a function of the density[4]. Alternatively, this can be a particular behaviour of the EAGLE simulation and further research will need to be done (ongoing research seems to suggest it, but no clear conclusions have been found yet). As we do not have enough observational data to cross-check the validity of this statement, the subse-

quent derived assumptions will be done pressuming no dependence with stellar-mass, but cannot be taken as more than simple lucubrations, as a bigger sample of observations is needed to verify this affirmation.

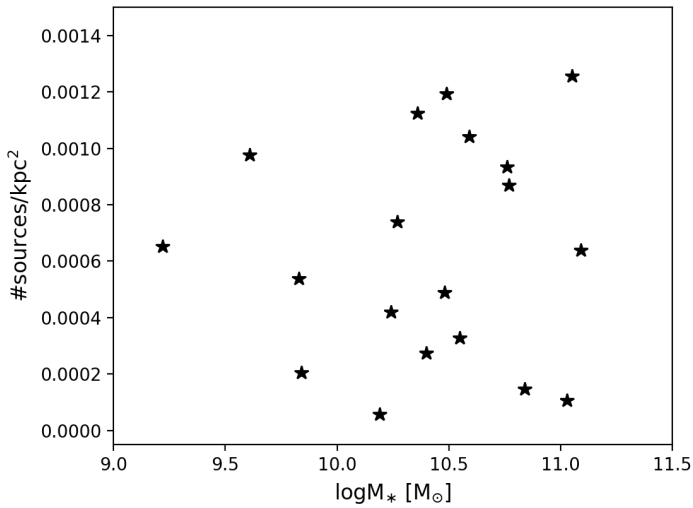


Figure 5.20: *Total number of neighbours per circular area of radius $r_{max} \approx 487kpc$ as a function of the stellar-mass of the centre. Results are computed using EAGLE simulation, without restriction on the galaxy activity.*

Regarding the behaviour as a function of source-type, we do not have a large enough sample of AGN to make any assumption with respect to SFG, but with the ones available, there is no obvious dependence between the number of neighbours and the type of central source. An attempt of also studying radio-selected AGN and no-radio AGN was unsuccessful, due to the small number of both types in our sample, as we can see in Fig 5.21, where all we can say is that the second ones tend to have smaller stellar-masses.

The last property analysed came across only after the classification of EAGLE sources regarding its galaxy activity. Those sources named as *errors* happened to have a systematically larger number of neighbours than any of the other sources with star-forming activity (both composites and SFG). This property was already explored in larger scales, where star-forming galaxies were more predominant at low densities while inactive (in the sense of low SFR, instead of related with AGN activity) lived in high density environments[4]. This can be seen in Fig. 5.22, where galaxies classified as composite and SFG are plotted together with its equivalent in *error* (what we described as inactive). A better understanding of what these *error* galaxies are, as well as observed counterparts, are needed to dig deeper in this claim.

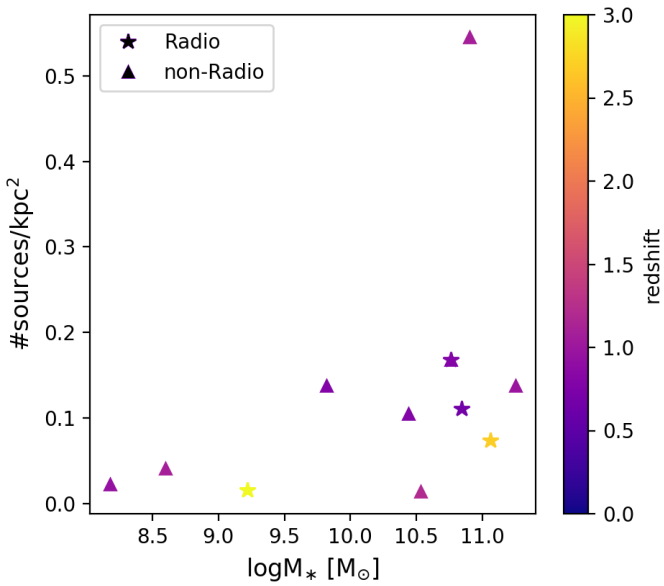


Figure 5.21: *Total number of neighbours per circular area of radius $r_{max} \approx 487kpc$ as a function of the stellar-mass of the centre, for both radio-AGN and no radio-AGN. Results are compute using the observational data. It also shows the redshift of each source.*

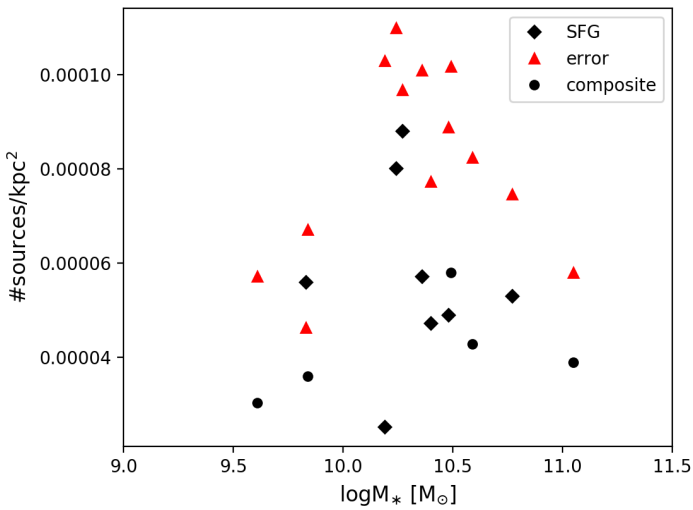


Figure 5.22: *Total number of neighbours per circular area of radius $r_{max} \approx 487kpc$ as a function of the stellar-mass of the centre. Results are computed using EAGLE simulation, specifying the type of source. It shows the behaviour of star-forming galaxies (SFG+composites) and that of the inactive galaxies (low SFR).*

6. Conclusions

The impact of spectroscopic redshift availability over using photometric redshifts for the study of galaxy-density environment has been translated into the possibility of reducing the scale of influence with respect to previous studies, and being able to perform a counting one-by-one of the galaxies surrounding a central source. This provides a way of accounting for the density evolution as a function of the distance to the centre.

Regarding the results on the environmental dependence of the different properties of galaxies studied (redshift, stellar-mass and galaxy activity), although promising, show that there is still a long way to go. The sample resulted to be not representative enough, and in order to achieve further progress, more centres of each type, bin mass and redshift are needed. Not only of radio-detected sources, but also other wavelengths if we want to compare its behaviour with no-radio emitters.

On the other hand, the election of a field such as COSMOS, with a rich multiwavelength catalogue, has resulted in a more favourable, multifaceted classification of galaxy activity and subsequently in a more trustworthy determination. Furthermore, it also allowed a better understanding of the origin of composite galaxies, through the analysis of the, apparently contradictory, results of the multiples diagrams.

With respect to the performance of EAGLE simulation, the obtained result are able to reproduce the observations at a really high detail, opening the possibility of carrying out further environment studies throughout this model. However, the dependence of environment with stellar-mass seems to contradict previous studies, and further analysis needs to be done in exploring this apparent limitation. A necessary test to carry out, as the scale of this research has decrease in relation with the work done until now, will consist on study multiple bins at similar redshift but in a wide interval of mass, and look for a possible relation.

Finally, future tests on EAGLE execution could include a study of the morphology of galaxies, or a match with the distribution of galaxies in the observed field of view (to especially account for particular overdensities), among others.

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Last but not least, my friends in Spain, who always tried to be close despite of being almost 2000km apart.

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